

Three Equivalent Routes to Estimating Power Spectra: Averaging, Multitaper, and Smoothing

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Abstract

Objective: Multitaper parameters have no counterpart in the two methods most users learn first, averaging the periodograms of segments and smoothing one periodogram across frequency. *Methods:* We write each estimator as a bank of tapers with weights and compare kernels and degrees of freedom. We score the estimators at a fixed resolution on simulated spectra, seizure EEG and sleep spindles, and with each tuned to its best setting. *Results:* The three methods are one estimator, built from one window by shifts in time, shifts in frequency, or orthogonal tapers. With all N Slepian tapers weighted by their eigenvalues, the multitaper estimate equals the periodogram averaged over a box and the average of periodograms of a long sliding sinc window. With fewer tapers it is a smoothed periodogram plus a cross term, which averages to zero for white noise and otherwise removes leakage. For every route the degrees of freedom are about twice the window length times the frequency resolution, so users choose the resolution and the noise follows. Where the spectrum is low, the untapered smoothed periodogram errs by 14 dB, the default multitaper estimate by 6 dB, and a tapered smoothed periodogram by 1 dB. On EEG the routes agree to a median of 0.6 to 0.9 dB. *Conclusion:* At equal resolution, a lightly tapered, smoothed periodogram matches an adaptively weighted multitaper estimate in leakage and accuracy. *Significance:* Users can choose time and frequency resolution from the features of their data and use whichever route their pipeline computes. *Keywords:* Electroencephalography, multitaper, periodogram, seizure, Slepian sequences, spectral estimation, Welch’s method.

1 Introduction

Three families of spectral estimator are in common use in neurophysiological signal analysis. The first averages periodograms. The record is cut into segments, a periodogram is computed for each, and the periodograms are averaged. This is Bartlett’s method [1], or Welch’s [2] when the segments are

tapered and overlap, and textbooks such as Kay’s [3] present it first. The second smooths a periodogram. One periodogram of the whole record is convolved across frequency with a box (Daniell [4]) or a Gaussian. This is the Blackman–Tukey estimator [5], and many analysis pipelines apply it to spectrograms without naming it. The third is Thomson’s multitaper method [6], which two tutorials for neuroscientists [7, 8] recommend because it controls leakage, reduces variance, and is optimal in a defined sense. Its two parameters, however, have no counterpart in the first two methods. A user who knows how to choose a segment length or a smoothing width has no obvious way to decide what time–bandwidth product to use, or whether to keep seven tapers or six. The tapers themselves, Slepian’s discrete prolate spheroidal sequences [9], are also unfamiliar to most users.

We show that the three methods are the same estimator (Fig. 1). Each is a quadratic form in the data, and a quadratic form can be written as a bank of tapers with weights. Averaging periodograms builds the bank by shifting one window along the record. Smoothing builds it by shifting one taper along the frequency axis. Thomson’s method builds it from orthogonal tapers. Each bank has an expected-value kernel and an equivalent number of degrees of freedom, so the three methods can be compared on the same terms. One of the correspondences is exact. The multitaper estimate with all N Slepian tapers, weighted by their eigenvalues, equals the periodogram averaged over a box of width R . This is equation (8.3) of Thomson’s paper [6], and neither tutorial mentions it. Averaging the periodograms of a sliding window is itself a form of smoothing [2, 10], so sliding a sinc window across the record produces the same estimate, and the Slepian tapers are the principal components of that sliding window. With the smaller number of tapers used in practice, the correspondence remains exact once a second term is included. The multitaper estimate is then a periodogram smoothed with a kernel built from the tapers, plus a cross term between different frequencies. The cross term is zero on average for white noise. For other spectra it removes the leakage that smoothing alone leaves in place.

These results have three practical consequences, which we measured over 300 realizations of a spectrum spanning 65 dB. Leakage survives smoothing and averaging. Where the spectrum is low, a box-smoothed periodogram errs by 14 dB without a taper and by 1.4 dB with one. A Hann taper has a third fewer degrees of freedom than the multitaper estimate at the same bandwidth ($\nu = 8.9$ against 14.0), which raises the error

^{*}This work was supported by the National Institutes of Health under Grants R01HL161253 and R01NS126282.

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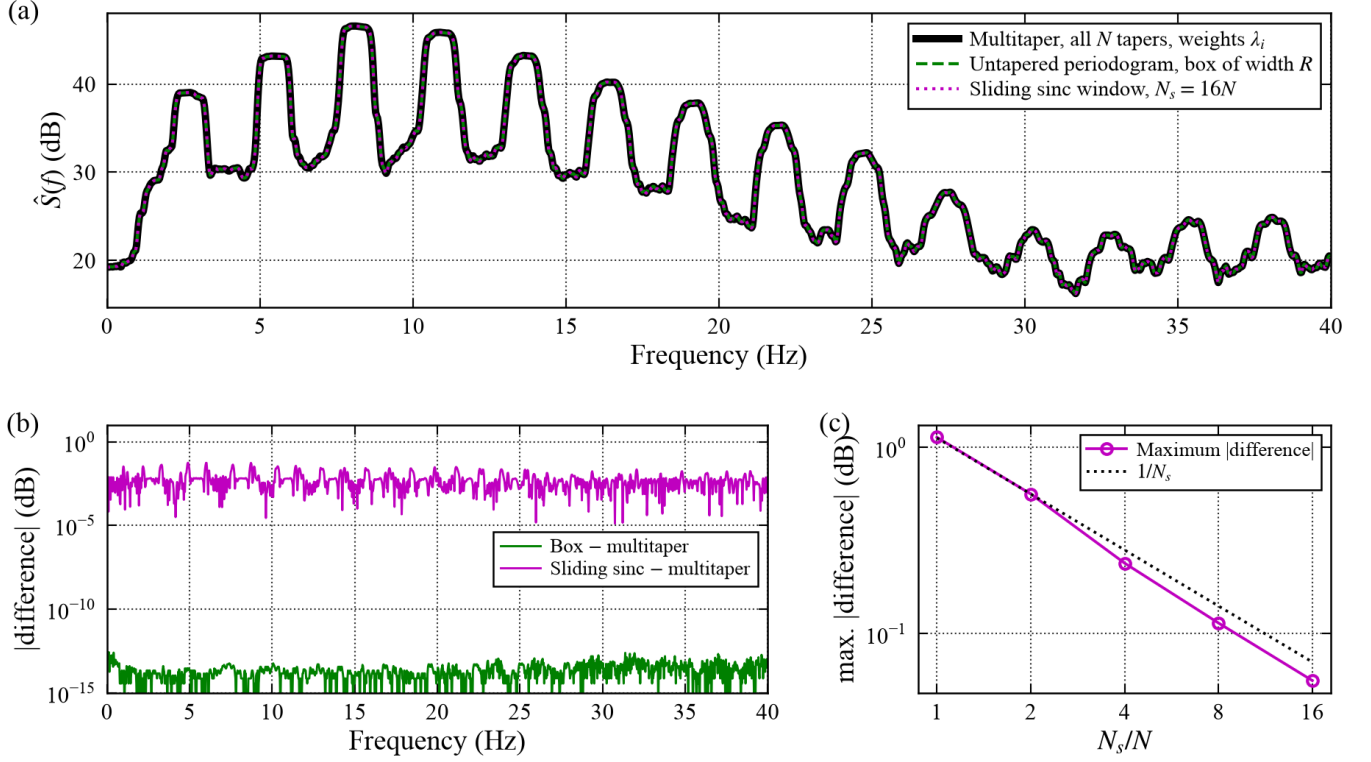


Figure 1: Three routes to one estimate of the power spectral density (PSD) of eight seconds of one channel of de-identified scalp EEG at 128 Hz from the authors’ archive ($N = 1024$; one epoch), with design resolution $R = 8/(N\Delta) = 1$ Hz ($\alpha = 4$). (a) The multitaper estimate with all N Slepian tapers weighted by their eigenvalues λ_i (thick black), the untapered periodogram smoothed with a box of width R (green, dashed), and the average of the periodograms of a sinc window of bandwidth R and length $N_s = 16N$ slid across the record one sample at a time (magenta, dotted). (b) Absolute differences from the multitaper estimate: the smoothed periodogram agrees to 3×10^{-13} dB and the sliding-window average to 0.06 dB. (c) The maximum difference of the sliding-window average against the window length N_s , with a $1/N_s$ guide line (dotted). The multitaper estimate here has $\nu = 17$ equivalent degrees of freedom, as defined in Section 2.2.

at the spectral peaks from 1.8 to 2.3 dB. A cosine taper applied only to the ends of the record avoids this loss ($\nu = 14.8$ with the same box, and 16.7 with a parabola of the same half-power width, whose base is wider). With default settings the multitaper estimate errs by 6 dB where the spectrum is low, and by 15 dB on records of 128 samples, while a tapered, smoothed periodogram errs by 1.0 to 1.4 dB. Adaptive weights remove this difference. When every route is tuned to its best setting, a tapered, smoothed periodogram is at least as accurate as the multitaper estimate in each of four test cases.

The paper makes six contributions. We prove that averaging periodograms, smoothing a periodogram and multitaper estimation are one estimator (Section 2.6). We derive the sliding-window form of Thomson’s identity and show that the Slepian sequences are the principal components of a sinc window slid across the record (Section 2.6.3, Appendix C). We write the multitaper estimate with any number of tapers as a smoothed periodogram plus a cross term, and give the share of its variance that the cross term carries (Section 2.6.5). We show that the variance cost of a single taper belongs to the taper and not to smoothing, that the best smoothing kernel is a parabola, and that for white noise the smoothed periodogram

has the smallest variance of all estimators with a given kernel (Sections 2.4.3 and 2.6.5). We compute exact kernels, leakage and degrees of freedom for eleven estimators at matched bandwidth, together with their errors over an ensemble (Tables 1 and 2). We show that the degrees of freedom of every route are about twice the product of the window length and the frequency resolution, and we give a recipe for each route that starts from the resolution required in time and in frequency (Sections 2.4 and 4.1). We test both on scalp EEG containing focal seizures and on sleep spindles (Sections 3.2 and 3.3).

The method is the same throughout. We write each estimator as a quadratic form, diagonalize it, and compare estimators through their kernels and eigen-weight ladders. Section 2.1 sets out the notation, which is that of Babadi and Brown [7]. Sections 2.2 and 2.3 introduce the kernel picture and the three classical remedies. Section 2.5 presents multitaper estimation in the usual way. Section 2.6 shows that the three methods are one and quantifies how the versions used in practice differ. Sections 4.1 and 3.2 give the recipes and the EEG demonstration. Proofs are in the appendices.

2 Methods

2.1 Notation

We follow the notation of Babadi and Brown [7], and every symbol used below is defined here or where it first appears. The symbol $:=$ marks a definition, $\mathbf{E}\{\cdot\}$ is the ensemble expectation, a prime denotes the transpose, and an asterisk denotes the complex conjugate, or for a vector the conjugate transpose. Let $x(t)$ be a zero-mean second-order stationary process sampled at the rate $F_s := 1/\Delta$, where Δ is the sampling interval, and let $x_k := x(k\Delta)$, $k = 0, 1, \dots, N-1$, be a record of N samples and duration $T := N\Delta$. Its autocovariance sequence is $s_k := \mathbf{E}\{x_i x_{i+k}\}$, and its power spectral density (PSD) is

$$S(f) := \Delta \sum_{k=-\infty}^{\infty} s_k e^{-i2\pi k f \Delta}, \quad |f| \leq \frac{1}{2\Delta}. \quad (1)$$

The Fourier transform of the record is

$$X(f) := \Delta \sum_{k=0}^{N-1} x_k e^{-i2\pi k f \Delta}, \quad (2)$$

and $x_f := [x_0, x_1 e^{-i2\pi f \Delta}, \dots, x_{N-1} e^{-i2\pi(N-1)f \Delta}]'$ is the record modulated to frequency f . A taper is a sequence h_k , $k = 0, \dots, N-1$, normalized to unit energy, $\sum_k h_k^2 = 1$, with transform $H(f) := \Delta \sum_k h_k e^{-i2\pi k f \Delta}$. The tapers of a set carry a superscript, $h_k^{(i)}$ and $H^{(i)}(f)$ for $i = 1, 2, \dots$, and $h^{(i)} := [h_0^{(i)}, \dots, h_{N-1}^{(i)}]'$ is the i th taper as a vector. An estimate of $S(f)$ is written $\hat{S}(f)$, with a superscript for the estimator: $\hat{S}^p$ is the periodogram, $\hat{S}^t$ a tapered periodogram, $\hat{S}^{(i)}$ the periodogram tapered with $h^{(i)}$, $\hat{S}^{\text{mt}}$ a multitaper estimate and $\hat{S}^{\text{sm}}$ a smoothed periodogram. A multitaper estimate uses L tapers and is designed for a spectral resolution R , the full width of the band $[-R/2, R/2]$ in which the tapers concentrate their energy. Its time-bandwidth product is [7, eq. (22)]

$$\alpha := \frac{N\Delta R}{2}. \quad (3)$$

We use R for the design resolution of every route. In numerical examples we give frequencies as $f\Delta$, in cycles per sample, and widths in units of $1/(N\Delta)$, so that $R = 8/(N\Delta)$ means $\alpha = 4$.

2.2 The Periodogram and the Kernel Picture

Every estimator in this paper returns the true spectrum blurred by a kernel, plus noise. For the periodogram the kernel is Fejér's and the noise is at its largest. The periodogram is

$$\hat{S}^p(f) := \frac{1}{N\Delta} |X(f)|^2 = \frac{\Delta}{N} \left| \sum_{k=0}^{N-1} x_k e^{-i2\pi k f \Delta} \right|^2, \quad (4)$$

the scaled squared magnitude of the discrete Fourier transform. Zero-padding before the fast Fourier transform (FFT) evaluates it on as fine a grid as needed. It has two well-known defects, both visible in Fig. 2(a, e).

Noise At any one frequency, the periodogram of a Gaussian signal is distributed as the true power spectral density (PSD) multiplied by a chi-square variable with two degrees of freedom, divided by two. Its standard deviation therefore equals its mean. A longer record does not reduce the noise, because it adds frequency points that are each as noisy as before.

Leakage The expected value of the periodogram is the true PSD S convolved with a kernel [7],

$$\mathbf{E}\{\hat{S}^p(f)\} = \int_{-1/2\Delta}^{1/2\Delta} N\Delta \mathcal{D}_N^2((f-f')\Delta) S(f') df', \quad (5)$$

where

$$\mathcal{D}_N^2(f\Delta) := \frac{\sin^2(N\pi f\Delta)}{N^2 \sin^2(\pi f\Delta)}. \quad (6)$$

The function $N\Delta \mathcal{D}_N^2(f\Delta)$ is Fejér's kernel, the squared transform of the rectangular window that truncated the record, divided by Δ . It has a main lobe about $2/(N\Delta)$ wide and side lobes that decay only as $1/f^2$. Neural spectra vary by tens of decibels across the band. The tails of the kernel then carry power from the strong part of the spectrum into the weak part, where the estimate is set by leakage instead of by the signal.

The kernel picture Equation (5) generalizes. Every estimator in this paper satisfies

$$\mathbf{E}\{\hat{S}(f)\} = \int_{-1/2\Delta}^{1/2\Delta} \mathcal{K}(f-f') S(f') df', \quad (7)$$

where $\mathcal{K}$ is the *expected-value kernel* of the estimator, which Babadi and Brown call its equivalent smoothing kernel [7], and the estimate fluctuates around this expected value. Three quantities characterize an estimator. Its resolution is the width of the main lobe of $\mathcal{K}$. Its leakage is the height of the skirts of $\mathcal{K}$. Its variance is reported as an equivalent number of degrees of freedom ν , meaning that the estimate behaves like $S\chi_\nu^2/\nu$. The relative standard error is then $\sqrt{2/\nu}$, or about $4.3\sqrt{2/\nu}$ dB, and an approximate 95% confidence interval runs from $\nu\hat{S}/\chi_{\nu,0.975}^2$ to $\nu\hat{S}/\chi_{\nu,0.025}^2$. The periodogram has the narrowest main lobe possible, high leakage, and $\nu = 2$. Each remedy below improves one of the three quantities at the expense of another. Appendix B gives the formulas used to compute the kernel and ν of each estimator exactly, and all numbers quoted below come from them.

2.3 Three Classical Remedies

2.3.1 Averaging the periodograms of segments

Bartlett's remedy for the noise [1] is to cut the record into M segments of N_s samples, compute a periodogram of each, and average them. Segments that do not overlap give independent estimates with $\nu = 2$ each, so the average has $\nu \approx 2M$. Resolution is lost in exchange. The kernel of each segment is the Fejér kernel of length N_s , whose main lobe is $2/(N_s\Delta) = 2M/(N\Delta)$ wide, and averaging does not narrow it. Welch [2] tapers each

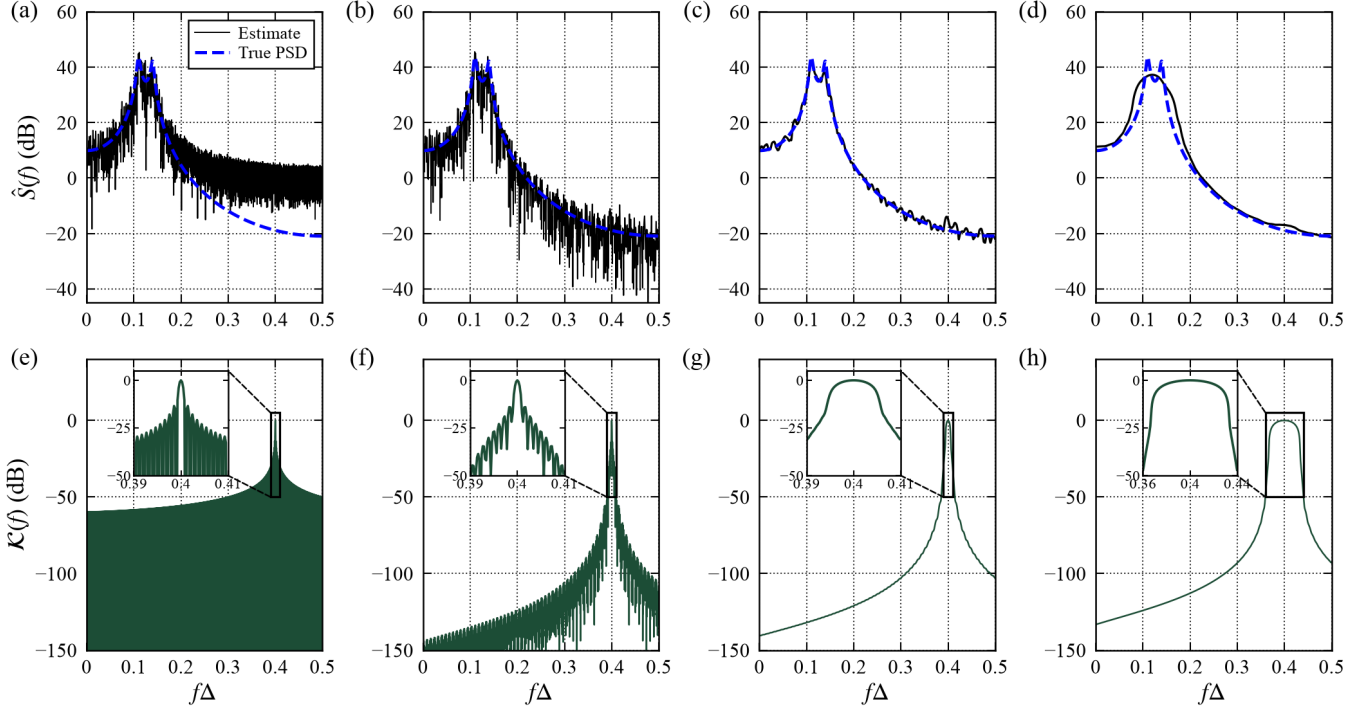


Figure 2: Every estimator is the true spectrum blurred by a kernel, plus noise. One realization ($N = 1024$) of an autoregressive process whose PSD (blue, dashed) has two peaks $7/(N\Delta)$ wide at half power and $30/(N\Delta)$ apart on a background spanning 65 dB. (a) Periodogram estimate. (b) Periodogram of the record after a 25% cosine taper. (c) The same, smoothed with a parabola whose half-power width $R = 8/(N\Delta)$ is that of a peak. (d) Smoothed with $R = 48/(N\Delta)$, which exceeds the separation of the peaks. The equivalent degrees of freedom are $\nu = 2, 2, 16.7$ and 98.4 . (e–h) The expected-value kernel of each estimator in dB relative to its peak, centered at $f\Delta = 0.4$. The zoomed-in view of the main lobe is shown on the top left. The half-power widths are $0.8, 1.2, 8.2$ and $47.8/(N\Delta)$, and $1.3\%, 0.02\%, 0.12\%$ and 65% of the kernel’s power lies outside $|f| \leq 8/(N\Delta)$, our measure of broadband leakage. The skirts of the Fejér kernel fall slowly, so power leaks from the peaks into the region 40 dB below them (a, e). The cosine taper lowers the skirts by 22 dB at $16/(N\Delta)$ and by 39 dB at $40/(N\Delta)$, and widens the main lobe by 15% (f). A kernel as wide as a peak is just right (g), and a wider kernel merges the peaks (h).

segment to control leakage and overlaps the segments by half their length to recover the data lost at the tapered ends. Hann segments at 50% overlap are nearly uncorrelated [2, 11], so ν is again close to twice the number of segments (13.4 for seven segments in Table 2). Whatever the overlap, the kernel of the estimator is that of the segment taper h of length N_s , $\mathcal{K}(f) = |H(f)|^2/\Delta$. The segment length therefore sets the bandwidth, and using more and shorter segments raises ν and widens the kernel.

2.3.2 Smoothing one periodogram

The second remedy computes one periodogram of the whole record and averages it over a window of frequencies. The window may be a box of width R [4], a Gaussian, or any non-negative kernel G of unit area, and the estimate is $\hat{S}^{\text{sm}}(f) := \int G(f - f') \hat{S}^{\text{p}}(f') df'$. Because expectation commutes with averaging, the kernel of the smoothed estimate is the kernel of the unsmoothed one convolved with the smoothing window, $\mathcal{K} = G * N\Delta \mathcal{D}_N^2$. The main lobe widens to about R , which the user sets directly. The noise falls, because periodogram values $1/(N\Delta)$ apart are nearly independent and a band of width R

pools about $N\Delta R = 2\alpha$ of them, giving $\nu \approx 2 \cdot 2\alpha$. The leakage does not change, because convolution with a non-negative window redistributes the skirts and cannot cancel them. This is the Blackman–Tukey estimator [5] computed in the frequency domain. Convolution of a spectrogram with a Gaussian along the frequency axis is the same operation. Section 2.6.3 shows that smoothing and averaging are the same remedy.

2.3.3 Tapering

Averaging and smoothing do not reduce leakage, which is why both are usually applied to tapered periodograms. A taper h_k that falls smoothly to zero at both ends of the record multiplies the data before the transform, which gives the tapered periodogram $\hat{S}^{\text{t}}(f) := \Delta |\sum_k h_k x_k e^{-i2\pi k f \Delta}|^2$. Its kernel is $|H(f)|^2/\Delta$ [7, eq. (14)]. We use a cosine (Tukey) taper, which is flat over the middle of the record and falls to zero as half a cosine at each end (Fig. 3a). With one eighth of the record tapered at each end, which we call a 25% cosine taper, the tails of the kernel decay as $1/f^6$ instead of $1/f^2$ beyond about $8/(N\Delta)$. They are 22 dB below those of the untapered kernel

at $16/(N\Delta)$ and 39 dB below at $40/(N\Delta)$, and the main lobe is 15% wider (Fig. 2f). The Hann window tapers the whole record. Its tails are a further 25 dB lower and its main lobe is 60% wider than that of the untapered kernel. Tapering does not reduce the variance, and a single tapered periodogram still has $\nu = 2$ (Fig. 2b). A taper also discards data, because the samples near the ends receive little weight. After smoothing, ν is divided by the variance-inflation factor $C_h := N \sum_k h_k^4$ of Percival and Walden [11, eq. (248a)], which is 1.95 for the Hann window and 1.15 for the 25% cosine taper at $N = 256$. The factor is asymptotic. It holds when the kernel spans many Fourier frequencies but a vanishing fraction of the band, and at $\alpha = 4$ the exact ratios of ν are 1.94 to 1.98 and 1.17. The taper sets how fast the skirts fall. The smoothing window, or the segment length, sets the width of the main lobe.

2.4 Choosing the Resolution

2.4.1 Time resolution, frequency resolution and noise

A user of a spectrogram makes two choices, the length $T = N\Delta$ of the window and the width of the kernel. Events closer together than T are merged in time, and peaks closer together than the resolution R , the half-power width of the kernel, are merged in frequency. The two resolutions fix the noise, because

$$\nu \approx 2TR = 4\alpha, \quad (8)$$

with T in seconds and R in hertz. For windows of 1 to 8 s and design resolutions of 1 to 4 Hz, the ratio of ν to $2TR$ is between 0.91 and 0.98 for the multitaper estimate of Section 2.5 with $L = 2\alpha - 1$ tapers and between 1.03 and 1.15 for recipe (b'). A user therefore chooses two of the three quantities, time resolution, frequency resolution and noise, and the third follows, by whichever route the estimate is computed. In EEG the choice is constrained by physiology. Sleep spindles last 0.5 to 2 s and lie between 11 and 16 Hz, and epileptiform discharges last less than a second. Windows of 1 to 2 s and kernels 1 to 2 Hz wide leave ν between 2 and 8. In this range the time-bandwidth product $\alpha = TR/2$ is 1 or 2, and the multitaper estimate has one to three tapers.

2.4.2 Choosing the bandwidth

Each remedy requires a resolution R , set by the segment length, by the width of the smoothing kernel, or by the time-bandwidth product α of (3). Fig. 3 shows how the choice affects a smoothed periodogram. A wider kernel pools more periodogram ordinates, so the noise falls as $4.3\sqrt{2/\nu}$ dB with ν roughly proportional to R (blue curve). A wider kernel also blurs the spectrum over a width R . A peak narrower than R is lowered and broadened, and a valley narrower than R is partly filled. This bias is zero for a flat spectrum and grows with the curvature of S on the scale of the kernel (red curve). Noise and bias add in quadrature, and their total has a broad minimum where the two are comparable.

The position of the minimum depends on the narrowest feature that must be preserved. The peaks in Fig. 2 are $7/(N\Delta)$

wide, so a kernel with $R = 8/(N\Delta)$, about the width of a peak, is the widest that does not flatten them. With this kernel the estimate has $\nu = 16.7$ and follows the true spectrum to within its own noise of about ± 1.5 dB. A narrower kernel gives a noisier estimate than necessary ($\nu = 2$, error 6 dB). A kernel wider than the separation of the peaks merges them ($R = 48/(N\Delta)$, $\nu = 98$), and although the estimate looks clean, its error over the peak region is 5.5 dB. The optimum also depends on the region that is scored. It is $R \approx 12/(N\Delta)$ over the whole band, most of which has no features, and $R \approx 8/(N\Delta)$ over the peaks. For real data the optimum is therefore not known in advance. We suggest the rule that Fig. 3 illustrates. Identify the narrowest feature of interest, for example an alpha peak one to two hertz wide, and set R to about its width. Data-driven choices of bandwidth exist for smoothed periodograms [12, 13] and for multitaper estimates [14]. They optimize a criterion over the whole band, so the same caveat applies.

2.4.3 The shape of the smoothing kernel

The best shape for the kernel can be derived to leading order, for a tapered periodogram at a frequency away from 0 and $1/(2\Delta)$, where the spectrum is smooth on the scale of the kernel and the kernel is much wider than $1/(N\Delta)$. There a non-negative kernel G of unit area gives the smoothed periodogram a bias of about $\frac{1}{2}S''(f) \int u^2 G(u) du$ and a relative variance of about $(C_h/N\Delta) \int G^2(u) du$. The constant C_h is the cost of the taper (Section 2.3.3). Each shape has a width at which its mean square error is smallest, and that smallest error is proportional to $(\int u^2 G)^{2/5} (\int G^2)^{4/5}$. Among non-negative kernels this quantity is least for the parabola

$$G(u) = \frac{3}{4b} \left(1 - \frac{u^2}{b^2}\right), \quad |u| \leq b, \quad (9)$$

which Priestley derived for spectral estimation [15] and Epanechnikov for density estimation [16]. Its lag window, $g_\tau = 3(\sin a - a \cos a)/a^3$ with $a = 2\pi b\tau\Delta$, is the quadratic-spectral kernel of Andrews [17, Theorem 2]. Its only parameter is the half-width b . Its half-power width is $\sqrt{2}b$, and the smoothed estimate has $\nu \approx (10/3)N\Delta b/C_h$. The shape matters little. Asymptotically, at their best widths, the box and the Gaussian have root-mean-square errors 3% and 2% larger than the parabola. Kernels with negative lobes can do better, and at a fixed width the parabola is not optimal. The width matters more. The best half-width is

$$b_{\text{opt}} = \left(\frac{15 C_h}{N\Delta}\right)^{1/5} \left|\frac{S(f)}{S''(f)}\right|^{2/5}. \quad (10)$$

It depends on the curvature of the spectrum, which changes from one frequency to the next and is not known in advance, and only weakly on the length of the record. For the alpha peak of the EEG-like spectrum of Section 2.7, which is 1.4 Hz wide, (10) gives a half-power width of 1.5 Hz in a 2-second record, with or without the 25% cosine taper. The width that minimizes the exact relative mean square error

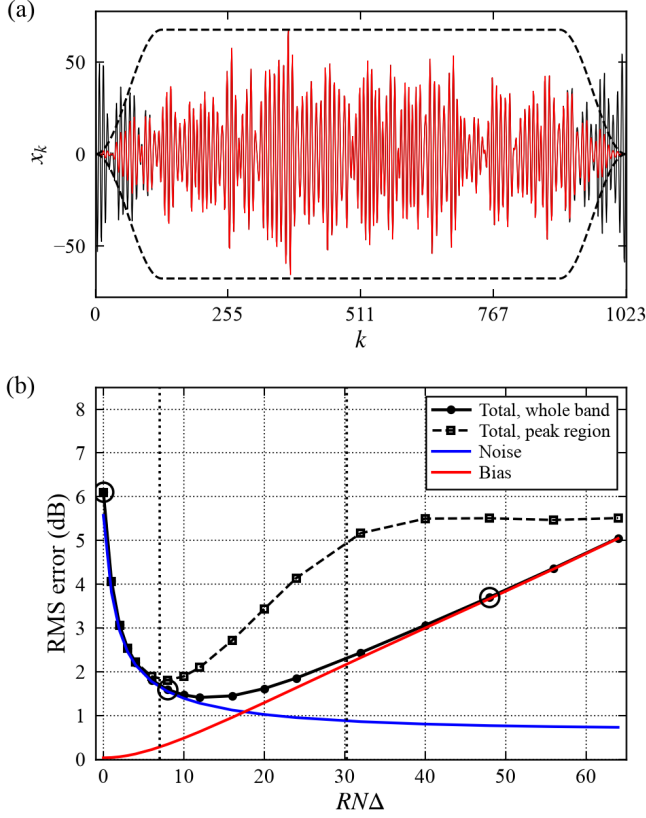


Figure 3: Choosing the resolution. (a) The record of Fig. 2, $N = 1024$ samples of the AR(4) process (black), the 25% cosine taper scaled to the record’s peak amplitude (dashed), and the tapered record (red). (b) Root-mean-square error over frequency, in dB, of the cosine-tapered periodogram smoothed with a parabola of half-power width R , against the true PSD, averaged over 300 independent realizations; the curves are ensemble averages and carry no error bars. The noise (the standard deviation of the estimate, blue) falls with R , the bias (blur by the kernel, red) grows, and their total over the whole band (solid black) has a broad minimum, which sits at smaller R when only the peak region, $0.09 \leq f\Delta \leq 0.16$, is scored (dashed). The circled points are panels (b), (c) and (d) of Fig. 2, and the dotted lines mark R equal to the width of a peak and to the separation of the peaks.

$\mathbf{E}\{(\hat{S}^{\text{sm}} - S)^2\}/S^2$ at 10 Hz is larger, 1.9 Hz without a taper and 2.0 Hz with it, because the peak is too narrow for the quadratic approximation. For the mean square error in decibels, in the chi-square approximation of Section 2.2, it is 1.65 and 1.74 Hz (demos/optimal_kernel.py, which also prints the 3% and 2%). All agree with the rule of setting the width of the kernel to about the width of the narrowest feature. Appendix D derives (9) and (10) by the calculus of variations.

2.5 Multitaper Estimation

Thomson’s method [6] achieves low leakage and low variance together, without shortening the record, by using several orthogonal tapers. The user chooses a resolution R . The tapers

are the orthonormal sequences of length N , $\sum_k h_k^{(i)} h_k^{(j)} = \delta_{ij}$, whose transforms are most concentrated in $[-R/2, R/2]$. The fraction of the energy of $H^{(i)}$ inside the band is the quadratic form $h^{(i)\prime} \Phi_R h^{(i)}$ in the $N \times N$ matrix [7, Sec. IV-A]

$$(\Phi_R)_{k,l} := \frac{\sin(\pi R \Delta(k-l))}{\pi(k-l)}, \quad (\Phi_R)_{k,k} := R\Delta, \quad (11)$$

so the tapers are its eigenvectors, $\Phi_R h^{(i)} = \lambda_i h^{(i)}$. These are Slepian’s sequences [9]. There are N of them, $h^{(1)}, \dots, h^{(N)}$, ordered by their concentrations $1 > \lambda_1 > \lambda_2 > \dots > \lambda_N > 0$, where λ_i is the fraction of the energy of the i th taper that lies inside the band. For $\alpha = 4$ the first six concentrations exceed 0.99, the seventh is $\lambda_7 = 0.94$, and the rest fall rapidly toward zero (Fig. 4d). Each taper gives an eigenspectrum

$$\hat{S}^{(i)}(f) := \Delta \left| \sum_{k=0}^{N-1} h_k^{(i)} x_k e^{-i2\pi k f \Delta} \right|^2, \quad (12)$$

and the estimate is the average of the first $L = 2\alpha - 1$ eigenspectra,

$$\hat{S}^{\text{mt}}(f) := \frac{1}{L} \sum_{i=1}^L \hat{S}^{(i)}(f). \quad (13)$$

The method works because its kernel is close to a box. Each taper has low leakage, so each eigenspectrum has little bias. The tapers are orthogonal, so the eigenspectra are nearly uncorrelated and the average of L of them has $v \approx 2L$. The kernel of (13) is the average of the L kernels $|H^{(i)}|^2/\Delta$ (Fig. 4b). The first is a single bump at the center of the band, the second has two bumps on either side of the center, and the third has three. Each kernel fills gaps left by the previous ones, and their sum is nearly flat over $[-R/2, R/2]$ and nearly zero outside it (Fig. 4c). Thomson’s optimality result is that no other set of L orthonormal tapers places more energy inside the band.

2.6 The Three Methods as One Estimator

2.6.1 Every estimator is a bank of tapers

Every estimator in this paper is a quadratic form in the data, and every quadratic form can be written as a bank of orthogonal tapers with weights. With the modulated record x_f of Section 2.1, $\hat{S}(f) = \Delta x_f^* Q x_f$ for a fixed real symmetric matrix Q [11, 18]. The eigendecomposition $Q = \sum_i c_i h^{(i)} h^{(i)\prime}$, with orthonormal eigenvectors $h^{(i)}$, gives

$$\hat{S}(f) = \sum_i c_i \Delta |h^{(i)\prime} x_f|^2 = \sum_i c_i \hat{S}^{(i)}(f), \quad (14)$$

which has the form of a multitaper estimate with tapers $h^{(i)}$ and weights c_i . The three methods differ in how they construct the bank (Fig. 5). Averaging periodograms places one window h at M offsets n and adds the periodograms with equal weight, so that $Q_{k,l} = \frac{1}{M} \sum_n h_{k-n} h_{l-n}$. The bank consists of *time-shifted* copies of one window, which are orthogonal only if the segments do not overlap.

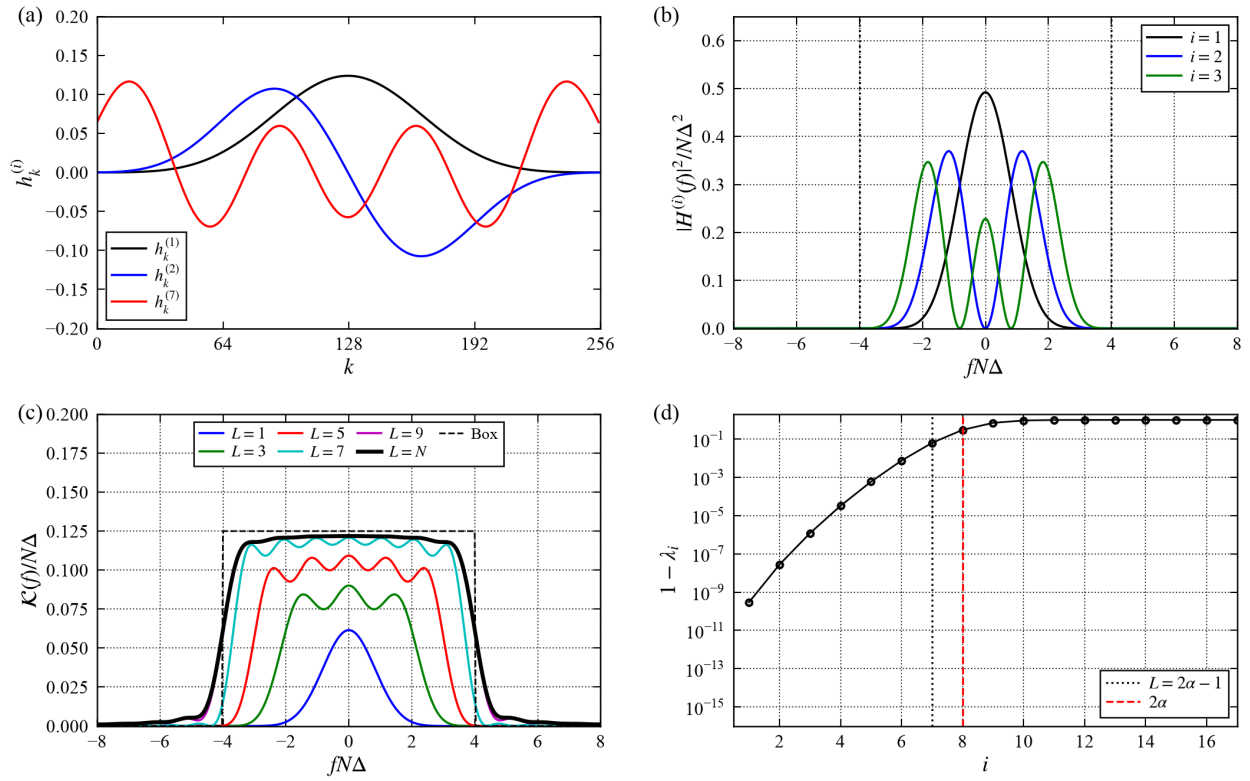


Figure 4: Why the multitaper kernel is a box ($N = 256$, $R = 8/(N\Delta)$). (a) The first (black), second (blue) and seventh (red) Slepian tapers. (b) The kernels $|H^{(i)}(f)|^2/\Delta$ of the first three, each filling the gaps left by the previous ones within the band $|f| \leq R/2$ (dotted lines). (c) The running sum $\sum_{i \leq L} \lambda_i |H^{(i)}(f)|^2 / 2\alpha\Delta$ as tapers are added. With all N tapers it is *exactly* the box of width R convolved with Fejér’s kernel, which is the identity (16) in kernel form and cannot be distinguished from the box itself (dashed) at this scale. With the usual $L = 2\alpha - 1 = 7$ it is the box with rounded shoulders and small ripple. (d) The concentration ladder: the fraction $1 - \lambda_i$ of each taper’s energy outside the band. Tapers beyond the 2α th carry only leakage.

Smoothing a tapered periodogram with a kernel G gives $\int G(f') \Delta |\sum_k h_k e^{i2\pi k f' \Delta} x_k e^{-i2\pi k f \Delta}|^2 df'$, a weighted sum of periodograms tapered with the modulated taper $h_k e^{i2\pi k f' \Delta}$. Here the bank consists of *frequency-shifted* copies of one taper, weighted by the kernel. Thomson’s method starts from an orthogonal bank.

The kernel and the degrees of freedom both follow from the bank. The kernel is $\mathcal{K} = \sum_i c_i |H^{(i)}|^2 / \Delta$. At an interior frequency the degrees of freedom are

$$\nu = \frac{2(\sum_i c_i)^2}{\sum_i c_i^2}, \quad (15)$$

which is Satterthwaite’s approximation [19] applied to the bank and agrees with the exact value to within 3% for the estimators we tested (Appendix B). The value is $2L$ for L equal weights and smaller when the weights are unequal. We call the ordered weights $c_1 \geq c_2 \geq \dots$ the *eigen-weight ladder*. Thomson’s ladder is flat by construction. The ladder of a smoothed single-taper periodogram slopes downward, because the modulated copies of one taper overlap. How much this costs depends on the taper and on the kernel. With a Hann window and a box the ladder is steep and short ($\nu = 8.9$). With a 25% cosine taper and a parabola it slopes gently over more

steps ($\nu = 16.7$; Fig. 5f). The ladder of Welch’s method with half-overlapping Hann segments is nearly flat, and the cost of that method appears in the shape of its kernel.

2.6.2 Thomson’s identity

Suppose that all N tapers are kept and each eigenspectrum is weighted by its concentration. Then

$$\frac{1}{2\alpha} \sum_{i=1}^N \lambda_i \hat{S}^{(i)}(f) = \frac{1}{R} \int_{f-R/2}^{f+R/2} \hat{S}^p(f') df' \quad (16)$$

The left side is the eigenvalue-weighted average of all N eigenspectra, whose weights sum to 2α . The right side is the periodogram averaged over a box of width R . This is Thomson’s equation (8.3) [6]. It holds for each record individually as well as in expectation, and we confirmed it to machine precision (Fig. 1).

The proof is brief. The Slepian tapers are the eigenvectors of Φ_R (11), and the λ_i are its eigenvalues. The entries depend only on the lag $k - l$, and as a function of lag they form a sinc, which is the inverse transform of the box. The matrix Φ_R therefore represents the box kernel, and $\Phi_R = \sum_i \lambda_i h^{(i)} h^{(i) \prime}$ is its decomposition into orthogonal components. Evaluating

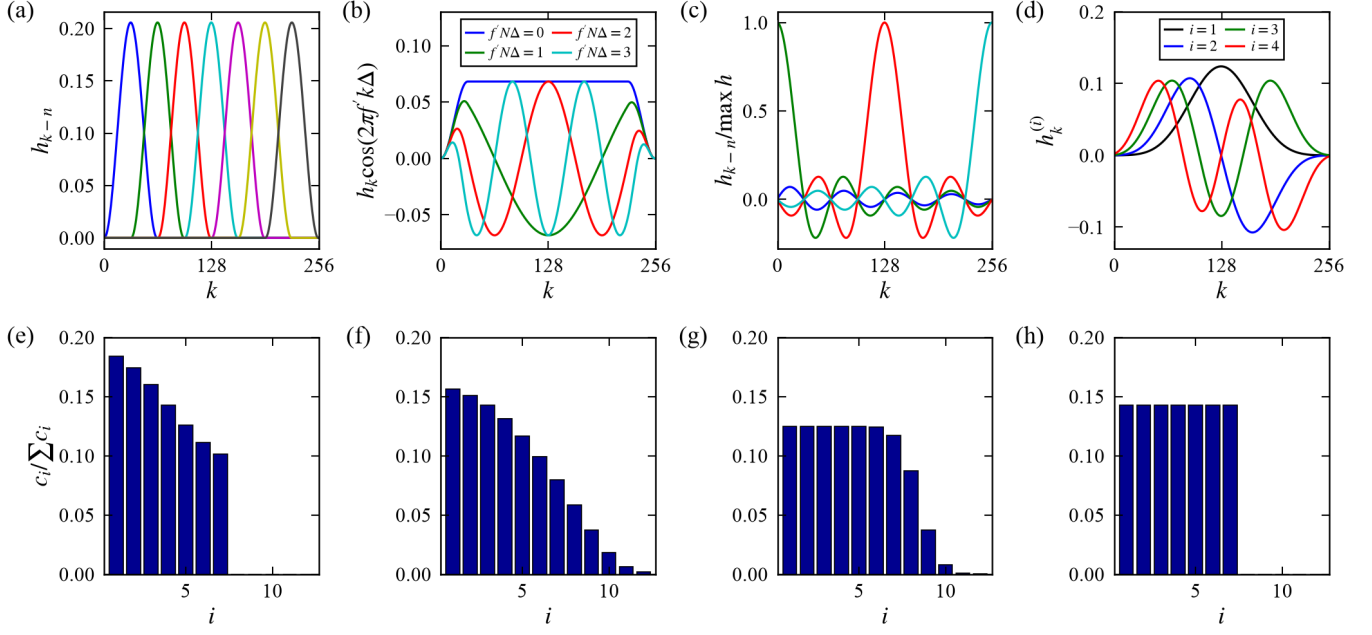


Figure 5: Three ways to build a bank of tapers from one window, and the reference bank ($N = 256$, $R = 8/(N\Delta)$). (a–d) The bank as constructed. (a) Welch’s method with Hann segments of length $N/4$ at 50% overlap: seven time-shifted copies of one window. (b) A periodogram tapered with a 25% cosine taper and smoothed with a parabola of half-power width R : frequency-shifted copies of one taper, four of a continuum. (c) A sinc window of bandwidth R slid across the record one sample at a time, four positions shown; in the limit of a long window this is exactly the untapered periodogram smoothed with a box, and exactly Thomson’s estimate with all N tapers weighted by λ_i . (d) Thomson’s estimate with $L = 7$ equally weighted Slepian tapers, the first four shown. (e–h) The eigen-weight ladder c_i of each, normalized to unit sum, whose flatness sets $\nu = 2(\sum_i c_i)^2 / \sum_i c_i^2$ by (15): $\nu = 13.4, 16.7, 17.2$ and 14.0 . The eigen-tapers of (c) are the Slepian sequences themselves. Fig. S4 adds the eigen-tapers and kernels.

$\Delta x_f^* \Phi_R x_f$ through the eigenvectors gives 2α times the left side of (16), and evaluating it through the lags gives 2α times the right side (Appendix A). In the terms of Section 2.6.1, the Slepian tapers are the eigen-tapers of the box-smoothed periodogram. Their concentrations form its ladder, which is flat near one for $i \leq 2\alpha$ and then falls to zero (Fig. 5g).

2.6.3 Averaging as smoothing

The matrix Φ_R also arises from averaging. Consider a window h , not normalized, that slides across the record one sample at a time and is allowed to extend beyond both ends of the zero-padded record. The periodograms of the windowed data are summed over all positions. Appendix C shows that the matrix Q of this estimator has entries $r_h(k-l)$, where $r_h(\tau) := \sum_k h_k h_{k+\tau}$ is the autocorrelation of the window. Averaging the periodograms of a sliding window is therefore exactly the same as smoothing the untapered periodogram with $|H|^2/\Delta$. For a sinc window $h_k = R\Delta \text{sinc}(R\Delta k)$, with $\text{sinc}(u) := \sin(\pi u)/(\pi u)$, the autocorrelation is $R\Delta \text{sinc}(R\Delta \tau)$, which is the lag sequence of

Φ_R , so that

$$\begin{aligned} \Delta \sum_n \left| \sum_k h_{k-n} x_k e^{-i2\pi k f \Delta} \right|^2 &= \Delta x_f^* \Phi_R x_f \\ &= \sum_{i=1}^N \lambda_i \hat{S}^{(i)}(f), \end{aligned} \quad (17)$$

where n runs over the offsets of the sinc window. The equality is exact for an infinitely long window. For a window of N_s samples the relative error in Q is 3.5% at $N_s = 2N$, 1.5% at $4N$ and 0.17% at $32N$, and it decreases as $1/N_s$ once N_s exceeds $2N$ (Appendix C). The matrix Φ_R , with entries $\sum_n h_{k-n} h_{l-n}$, is the scatter matrix of the shifted windows. Its eigenvectors, the Slepian sequences, are therefore the principal components of the sliding sinc window, and Thomson’s estimator retains the leading components.

2.6.4 The estimators used in practice

The standard estimate (13) differs from the left side of (16) in two ways. It uses equal weights, which changes little because $\lambda_i \approx 1$ for $i \leq L$. It also stops at $L = 2\alpha - 1$ tapers, and this

is the difference that matters. Rearranging (16) gives

$$\frac{L}{2\alpha} \hat{S}^{\text{mt}}(f) \approx \underbrace{\frac{1}{R} \int_{f-R/2}^{f+R/2} \hat{S}^{\text{p}}(f') df'}_{\text{box-smoothed periodogram}} - \underbrace{\frac{1}{2\alpha} \sum_{i>L} \lambda_i \hat{S}^{(i)}(f)}_{\text{leaky remainder}}. \quad (18)$$

The factor $L/2\alpha$, which is $7/8$ for $\alpha = 4$, arises because the standard estimate averages over L tapers while the weights in (16) sum to 2α . The eigenspectra that are dropped belong to tapers whose energy lies almost entirely outside $[-R/2, R/2]$, so each contributes at frequency f only power that has leaked in from other frequencies. The multitaper estimate is thus the box-smoothed periodogram, or equivalently the sliding-sinc average, with this leakage subtracted. Thomson noted that the two agree “only for white spectra” [6]. Stoica and Moses obtain the Slepian estimator from the smoothed periodogram in the same way, by replacing Φ_R with its L leading eigenvectors [20].

2.6.5 Smoothing before squaring, or after

Multiplication in time is convolution in frequency. The transform of the tapered record $h_k^{(i)} x_k$ is therefore the transform X of the record convolved with the transform $H^{(i)}$ of the taper, and the multitaper estimate is

$$\hat{S}^{\text{mt}}(f) = \frac{1}{L\Delta} \sum_{i=1}^L \left| \int_{-1/2\Delta}^{1/2\Delta} H^{(i)}(f-f') X(f') df' \right|^2. \quad (19)$$

It can be computed from one Fourier transform of the untapered record. It smooths the complex transform in L ways, squares each result, and averages. A smoothed periodogram performs these operations in the other order. It squares first and smooths afterwards,

$$\hat{S}^{\text{sm}}(f) = \frac{1}{N\Delta} \int_{-1/2\Delta}^{1/2\Delta} G(f-f') |X(f')|^2 df'. \quad (20)$$

The two orders differ because squaring discards the phase of X . The square in (19) expands into a double integral over products $X(f')X^*(f'')$ of the transform at pairs of frequencies, and (20) keeps only the terms with $f' = f''$. For a stationary process the products at different frequencies are close to zero on average and are not zero in any one record. This is why the two estimators can have the same expected value, and why their estimates of one record differ.

The two parts can be written out. Let $X_j := X(f_j)$ be the transform at the Fourier frequencies $f_j := j/(N\Delta)$, $j = 0, \dots, N-1$, and $\hat{S}_j^{\text{p}} := |X_j|^2/(N\Delta)$ the periodogram there. Then $\int H^{(i)}(f-f')X(f')df' = \frac{1}{N\Delta} \sum_j H^{(i)}(f-f_j)X_j$, and

expanding the square in (19) gives

$$\hat{S}^{\text{mt}}(f) = \frac{1}{N\Delta} \sum_j \mathcal{K}_L(f-f_j) \hat{S}_j^{\text{p}} + \frac{1}{(N\Delta)^2} \sum_{j \neq l} B_L(f-f_j, f-f_l) X_j X_l^*, \quad (21)$$

with $B_L(a, b) := \frac{1}{L\Delta} \sum_{i=1}^L H^{(i)}(a)H^{(i)*}(b)$ and $\mathcal{K}_L(a) := B_L(a, a)$, the kernel of (13). The first term is the periodogram smoothed with $\mathcal{K}_L$, the box with rounded shoulders of Fig. 4c, which is built from the L tapers. The second is a cross term between different frequencies. For white noise the cross term has zero mean and is uncorrelated with the first term, so the variances of the two terms add. At a Fourier frequency the cross term carries the fraction

$$d^2 = 1 - \frac{L}{(N\Delta)^2} \sum_{j=0}^{N-1} \mathcal{K}_L(f_j)^2 \quad (22)$$

of the variance of the multitaper estimate. The first term alone therefore has $\nu = 2L/(1-d^2)$, and its correlation with the multitaper estimate is $(1-d^2)^{1/2}$. For $\alpha = 4$ and $L = 7$ these are $d^2 = 0.08$, $\nu = 15.2$ and 0.96 . For $\alpha = 2$ and $L = 3$ they are 0.12 , 6.9 and 0.94 . For a coloured spectrum the mean of the cross term is not zero. It is negative where the spectrum is low, because the cross term subtracts the power that the first term admits through the side lobes of the Fejér kernel. On the AR(4) process of Section 2.7 the first term alone is about 20 dB too high in that region. The tapers act through the cross term (demos/frequency_domain_multitaper.py). McCloud, Scharf and Mullis reached the approximate form of this relation, without the cross term, from a low-rank factorization of Φ_R [21].

In terms of the matrix Q , an estimator has the form (20) exactly when Q is constant along each of its diagonals, that is, when Q is a Toeplitz matrix. The matrix Φ_R of Thomson’s identity is of this kind, which is why the estimate with all N tapers is a smoothed periodogram. The matrix of the L -taper estimate is not. The Toeplitz matrix closest to it is obtained by averaging each of its diagonals. It lies at a relative distance of 0.20 for $\alpha = 4$ and $L = 7$, at any N , and at 0.27 for the $\alpha = 2$, $L = 3$ of our EEG analysis.

For sine tapers [18] the convolution in (19) has only two terms, and the estimate is

$$\hat{S}(f) = \frac{1}{2L(N+1)\Delta} \sum_{i=1}^L |X(f-\delta_i) - X(f+\delta_i)|^2, \quad (23)$$

with $\delta_i := i/((2N+2)\Delta)$ and the time origin of X one sample before the record. It is a multitaper estimate computed from the FFT of the untapered record, by differencing before squaring.

We asked whether a smoothed periodogram could nevertheless reproduce the estimate used in practice. For the expected value the answer is yes. The kernel of a quadratic estimator is the transform $\mathcal{K}(f) = \Delta \sum_{\tau} q_{\tau} e^{-i2\pi\tau f\Delta}$ of the lag sums

$q_\tau := \sum_k Q_{k,k+\tau}$ of its matrix (Appendix B). For a tapered periodogram smoothed with the lag window g_τ , whose transform $G(f) = \Delta \sum_\tau g_\tau e^{-i2\pi\tau f\Delta}$ is the smoothing kernel, the lag sums are $g_\tau r_h(\tau)$, where r_h is the autocorrelation of the taper. The lag window

$$g_\tau = \frac{q_\tau}{r_h(\tau)}, \quad q_\tau = \frac{1}{L} \sum_{i=1}^L \sum_k h_k^{(i)} h_{k+\tau}^{(i)}, \quad (24)$$

therefore gives the smoothed periodogram the kernel of the L -taper estimate, exactly when $r_h(\tau) \neq 0$ at every lag where $q_\tau \neq 0$, as it is without a taper. Without a taper the estimator is the closest Toeplitz matrix described above. The symmetric Hann and cosine tapers vanish at their end samples, so the last two or three lags are not matched, and for $N = 256$ the kernels then differ by less than 3×10^{-4} of their peak. In the frequency domain, the smoothing kernel is the multitaper kernel $\mathcal{K}_L$ deconvolved by the kernel $|H|^2/\Delta$ of the taper. The deconvolution is ill-conditioned at the lags where r_h is small, so the lag window is applied in the lag domain. With an exact match the two estimators have the same bias for every spectrum.

They do not have the same variance. For real Gaussian white noise of unit variance, the variance of $x_f^* Q x_f$ is $\sum_{|\tau| < N} (1 + \cos 4\pi\tau f\Delta) e_\tau$ with $e_\tau := \sum_k Q_{k,k+\tau}^2$, and by the Cauchy–Schwarz inequality $e_\tau \geq q_\tau^2/(N - |\tau|)$, with equality when Q is constant along the diagonal at lag τ . For white noise, among all quadratic estimators with a given kernel, the smoothed periodogram without a taper therefore has the smallest variance at every frequency. For $N = 256$ and $\alpha = 4$, with the kernel of the L -taper estimate, it has $\nu = 14.6$ for $L = 7$ and $\nu = 10.9$ for $L = 4$. The multitaper estimates have $\nu = 14$ and $\nu = 8$. The result does not extend to coloured spectra. On the AR(4) process of Section 2.7, above $f\Delta = 0.2$, the smoothed periodogram with the $L = 7$ kernel has more than ten times the variance of the multitaper estimate. The matched kernel has negative lobes, and most of the eigen-weights of this estimator are negative (together about 3% of the positive total). It has no protection against leakage.

The two are therefore different estimators, and their estimates of the same record differ. For $N = 256$, $\alpha = 4$ and $L = 7$, the kernel-matched periodogram without a taper lies at a relative distance of 0.20 from the multitaper matrix. The closest pair of taper and lag window, found by alternating least squares, lies at 0.15, and its taper is nearly rectangular. On 1000 simulated records of the EEG-like spectrum of Section 2.7, the estimates of this pair differ from the multitaper estimates of the same records by 0.3 dB (root mean square) and the kernel-matched estimates by 0.6 dB. The error of each estimator is 2.0 dB. On the AR(4) process, which spans 65 dB, the differences are 1.1 and 4.0 dB. The second value is computed over the two thirds of the kernel-matched estimates that are positive, because the other third are negative and have no value in decibels. Its lag window cancels the leakage of the untapered periodogram on average and not in each record. Hansson-Sandsten posed the same question for Welch’s method [22], and our search over segment windows

came no closer than 0.27 (demos/fit_single_window.py, demos/kernel_matched.py). A single window can therefore reproduce the multitaper estimate closely when the dynamic range is modest, and cannot when it is large. In that case the other routes must taper before they smooth or average.

The taper can be built from the Slepian tapers as well. The diagonal of the multitaper matrix is $\frac{1}{L} \sum_i (h_k^{(i)})^2$, so the taper $h_k = (\frac{1}{L} \sum_i (h_k^{(i)})^2)^{1/2}$ with the lag window (24) reproduces the diagonal and the kernel together. For $N = 256$, $\alpha = 4$ and $L = 7$ this estimator lies at 0.16 from the multitaper matrix. Its estimates differ from the multitaper estimates of the same records by 0.3 dB on the EEG-like spectrum and by 0.8 dB on the AR(4) process. None of them is negative, and its error equals that of the multitaper estimate on both processes (demos/rounded_kernel.py).

2.7 Simulations

We compared the estimators on a fourth-order autoregressive process, AR(4), that is a standard test of leakage [11]. Its PSD has two peaks $7/(N\Delta)$ wide at half power and $30/(N\Delta)$ apart for $N = 1024$, on a background that spans 65 dB. We generated 300 realizations of $N = 1024$ samples and estimated the PSD of each with a design resolution of $R = 8/(N\Delta)$, that is $\alpha = 4$. The error of an estimate is the absolute difference in dB between the estimate and the true PSD. We report the median error over the whole band, the root-mean-square error over the peak region ($0.09 \leq f\Delta \leq 0.16$), and the median error where the spectrum is low ($f\Delta > 0.3$). Kernel properties and degrees of freedom were computed exactly from the quadratic form of each estimator (Appendix B), for $N = 256$ and $R = 8/(N\Delta)$ unless stated otherwise. The same comparison for $N = 256$ and $N = 128$ is in the repository. In the figures the smoothing route is recipe (b’) of Section 4.1, a 25% cosine taper followed by a parabola whose half-power width is R . The tables also give the Hann window followed by a box, which is the most familiar choice.

As a check that no route is favoured by the choice of bandwidth, we also tuned each family. The families were multitaper estimates with Slepian tapers, with equal or with adaptive weights, for time–bandwidth products from 1 to 12 in steps of 0.5 (from 1.5 with adaptive weights) and every number of tapers; sine tapers [18] with equal weights; one taper (Hann, a cosine taper of 10%, 25% or 50%, or one Slepian taper) followed by a box or a parabola of 17 widths; two to five Slepian tapers followed by a box; and Welch’s method with Hann segments of 12 lengths. For each family we shortlisted the five settings with the smallest error predicted from the exact mean and variance of the estimator (Appendix B), scored them on 1000 simulated records, and kept the best. The adaptively weighted estimates are not quadratic and were scored on 500 records at every setting. The error is the root mean square, over records and frequencies, of the difference in dB between the estimate and the true PSD. The test cases were the AR(4) process with $N = 256$ and $N = 1024$, and an EEG-like spectrum sampled at 200 Hz with $N = 400$ and $N = 1024$. The EEG-like spectrum has a background that falls as $1/f^2$ above

3 Hz, an alpha peak at 10 Hz that is 1.4 Hz wide, and a noise floor. It spans 30 dB and was scored from 1 to 40 Hz. The tuning uses the true spectrum, so it shows what each route can achieve and is not a procedure for choosing settings.

2.8 EEG Data and Settings

Fig. 8 applies the three practical estimators to a ten-minute scalp EEG clip that contains a focal seizure, and Table 3 adds a second clip that contains two. Clinicians read spectrograms of recordings like these. The clips are de-identified recordings from a seizure-annotation set (ELROND) drawn from the Brain Data Science Platform. They were sampled at 200 Hz on 20 channels, and one of the authors (M.B.W.) marked the seizure onsets and offsets. Retrospective analysis of the EEG data was conducted under Institutional Review Board protocols at Stanford University (#83833), Beth Israel Deaconess Medical Center (#2016P000058) and Massachusetts General Hospital (#2013P001024), with a waiver of consent. For each clip we show the channel with the largest ictal rise in power between 2 and 20 Hz. The features to preserve are the harmonics of the ictal rhythm. At the ictal instant of Fig. 8 the fundamental is at 7.7 Hz and the harmonics are 7.4 Hz apart. On the unsmoothed periodogram the peaks are 0.5 to 0.7 Hz wide. This is the resolution of a 2-second window, 0.5 Hz, and an upper bound on the width of the rhythm itself. In the second clip the harmonic spacing falls to about 4 Hz as the seizure slows. Following the rule of Section 2.4.2, the bandwidth should be well below the harmonic spacing and otherwise as wide as possible. We set $R = 2$ Hz in 2-second windows. The multitaper estimate uses $\alpha = 2$, $L = 3$ and equal weights, the smoothed estimate uses recipe (b'), a 25% cosine taper with a parabola whose half-power width is 2 Hz, and Welch's method uses 0.7-second Hann segments. Table 3 also gives the Hann taper with a box of width 2 Hz.

Fig. 9 uses 40 s of stage N2 sleep from a de-identified overnight polysomnogram from the same platform, sampled at 200 Hz, channel C4 referred to the left mastoid. The stage was scored by a technologist and agreed with automated staging. We marked spindles where the envelope of the signal between 11 and 16 Hz exceeded twice its median for 0.5 to 3 s. The spectrograms use windows of 1, 2 and 4 s stepped by 0.1 s, with design resolutions R of 2, 2 and 1 Hz.

3 Results

3.1 Simulations

Fig. 6 shows the estimators used in practice on the record of Fig. 2. Table 1 gives their errors over 300 realizations in three regions of the spectrum, Table 2 gives their exact kernel properties, and Fig. 7 shows them on EEG. The main findings are as follows.

Leakage survives smoothing and averaging The box-smoothed untapered periodogram has the intended resolution

and the most degrees of freedom. Where the spectrum is low it errs by 13.7 dB (Table 1), because the leakage in the Fejér tails is averaged and remains in the estimate, as (18) predicts. A tapered periodogram smoothed to the same width errs by 1.0 dB in that region with the cosine taper and by 1.4 dB with the Hann window. The taper controls the skirts of the kernel, and the box or segment length controls the width of its main lobe. Neither can substitute for the other. Prewhitening is the other classical remedy for a large dynamic range [5, 11], and it can be combined with any of the three routes.

A heavy taper costs degrees of freedom At the spectral peaks leakage is negligible and the errors follow ν . The multitaper estimate, with $\nu = 14.0$, errs by 1.8 dB. The Hann-tapered, box-smoothed periodogram has $\nu = 8.9$, a third fewer, because its ladder slopes, and it errs by 2.3 dB. The corresponding relative standard errors $\sqrt{2/\nu}$ are 38% and 47%. The cost can be paid in resolution instead. If the box is widened until the Hann route reaches $\nu = 14$, its kernel is $13/(N\Delta)$ wide, compared with $7.4/(N\Delta)$ for Thomson's. The deficit decreases as N grows, because the broadening caused by a single taper scales as $1/(N\Delta)$ and R does not [23].

The cost belongs to the Hann window The Hann window gives most of the record little weight, and a lighter taper recovers the loss. The 25% cosine taper followed by a parabola of the same half-power width has $\nu = 16.7$. With the box in place of the parabola it has $\nu = 14.8$, which is still more than the multitaper estimate. The parabola adds the rest because its base is wider than that of a box of the same half-power width. It errs by 1.7 dB at the peaks and by 1.0 dB where the spectrum is low, which is less than any multitaper estimate in Table 1. Its skirts are higher than those of the Hann route and lower than those of the multitaper estimate: beyond $1.5R$ the three kernels are below -38 , -66 and -31 dB. The price is the shape of the kernel. A parabola has no flat top and its base is 1.4 times its half-power width, so two equal tones are separated at $10.0/(N\Delta)$, compared with $9.0/(N\Delta)$ for the Hann-tapered box.

The last Slepian taper leaks For $\alpha = 4$ the seventh taper has $\lambda_7 = 0.94$, so 6% of its energy lies outside the band. This taper sets the skirt of the multitaper kernel at -22 dB (Fig. 6f), and the kernel is still at -31 dB beyond $1.5R$, where that of the smoothed periodogram is below -38 dB (Fig. 6g). With equal weights and $L = 2\alpha - 1$, the multitaper estimate errs by 6.2 dB where the spectrum is low, compared with 1.0 dB for the smoothed periodogram. On records of 128 samples the errors are 15 dB and 1.2 dB. Dropping the last taper reduces the multitaper error to 2.1 dB. Thomson's adaptive weights [6, 24] reduce it to 1.7 dB by down-weighting the leaky tapers at frequencies where the spectrum is far below its maximum. At those frequencies the adaptively weighted estimate has $\nu = 6.5$ instead of 14 (Fig. S2). Recent analysis recommends $L = 2\alpha - O(\log \alpha)$ for the same reason [25]. Over the whole band the adaptively weighted estimate and the

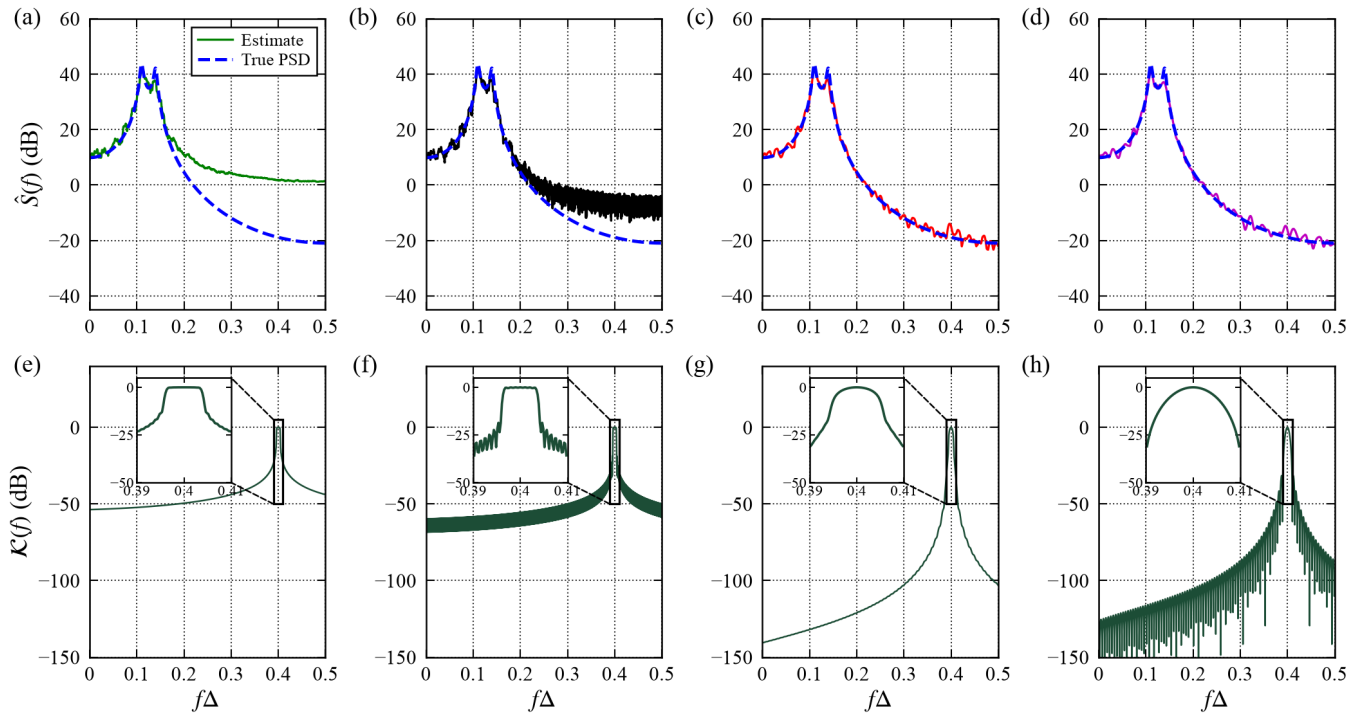


Figure 6: The estimators used in practice, applied to the record of Fig. 2 ($N = 1024$, $R = 8/(N\Delta)$; one realization; true PSD blue, dashed). (a) The untapered periodogram smoothed with a box, which is the exact estimate (16) ($\nu = 17.2$). (b) Thomson's estimate with $L = 7$ tapers ($\nu = 14.0$). (c) A cosine-tapered periodogram smoothed with a parabola of the same half-power width ($\nu = 16.7$). (d) Welch's method with nine Hann segments of 192 samples stepped by 104, chosen for about the same half-power width ($\nu = 17.6$). The four half-power widths are 8.2, 7.2, 8.2 and 7.8/($N\Delta$). This realization has more leakage than is typical. Its median absolute errors over the band are 11.6, 4.6, 0.9 and 0.8 dB, and the values over 300 realizations are 5.3, 2.3, 1.0 and 1.0 dB (Table 1). (e–h) The kernels, as in Fig. 2, with 1.4%, 0.22%, 0.12% and 0.48% of their power outside $|f| \leq R$. The untapered box (e) inherits the Fejér tails, and its estimate levels off 20 dB above the true spectrum. The seventh Slepian taper puts the skirt of the multitaper kernel at -22 dB (f), and that estimate levels off 10 dB above the true spectrum. The kernel of the cosine-tapered parabola (g) has no side lobes and falls below -38 dB beyond $1.5R$. Welch's kernel (h) has no flat top, and its side lobes are spaced at the scale of the peaks.

Table 1: Errors of the Estimators Used in Practice Over 300 Realizations

Estimator	ν	width	whole band	peaks	low spectrum
Untapered periodogram, then box	17.2	8.1	5.3	1.7	13.7
Multitaper, $L = 7$, equal weights	14.0	7.4	2.3	1.8	6.2
Multitaper, $L = 6$, equal weights	12.0	6.6	1.5	2.0	2.1
Multitaper, $L = 7$, adaptive weights	6.5–14.0	—	1.4	1.8	1.7
Two Slepian tapers ($\alpha = 2$), then box	13.3	8.1	1.3	1.9	1.6
Hann periodogram, then box	8.9	8.1	1.4	2.3	1.4
Cosine taper (25%), then parabola of half-power width $8/(N\Delta)$	16.7	8.1	1.0	1.7	1.0
Hann periodogram, then box of width $13/(N\Delta)$	14.0	13.1	1.2	2.3	1.1
Welch, nine Hann segments of 192 samples	17.6	7.6	1.0	1.8	1.0

AR(4) process, $N = 1024$, $R = 8/(N\Delta)$. Errors are absolute differences in dB between the estimate and the true PSD: the median over the whole band, the root mean square over the peak region ($0.09 \leq f\Delta \leq 0.16$), and the median where the spectrum is low ($f\Delta > 0.3$). Widths are half-power widths of the kernel in units of $1/(N\Delta)$. The adaptively weighted estimate has no single kernel, and its ν is 14.0 at the peaks and 6.5 for $f\Delta > 0.3$. Computed by `demos/everyday_comparison.py`, which also gives $N = 256$ and $N = 128$.

Hann-tapered, smoothed periodogram have the same error, 1.4 dB, and the cosine-tapered parabola errs by 1.0 dB. The Hann-tapered, smoothed periodogram therefore matches the adaptive estimator where the spectrum is low, and has a third fewer degrees of freedom where it is high.

Welch's method and hybrid estimators Welch's method with half-overlapping Hann segments, matched in half-power width to the other estimators, has the most degrees of freedom in Table 1 ($\nu = 17.6$) and, with the cosine-tapered parabola, the smallest errors. Its segments are nearly orthogonal, so its ladder is nearly flat. Its disadvantage is the shape of its kernel.

Table 2: Exact Degrees of Freedom and Kernel Properties at the Same Design Bandwidth

Estimator	ν	half-power width	leakage	peak side lobe
Ideal box (reference)	—	8.06	0	—
Multitaper, $L = 2\alpha - 1 = 7$, equal weights	14.0	7.44	0.0022	−22 dB
Multitaper, 7 sine tapers, equal weights	14.0	7.44	0.0008	−22 dB
Multitaper, all N tapers, weights λ_i	17.2	8.06	0.0138	−17 dB
Untapered periodogram, then box (= sliding sinc)	17.2	8.06	0.0138	−17 dB
Hann periodogram, then box	8.9	8.06	$< 10^{-4}$	−36 dB
Cosine taper (25%), then parabola	16.7	8.06	0.0013	−16 dB
Bohman periodogram, then box	7.3	8.06	$< 10^{-4}$	−31 dB
Two Slepian tapers ($\alpha = 2$), then box	13.3	8.06	0.0002	−34 dB
Hann periodogram, then Gaussian ($\sigma = 3/(N\Delta)$)	11.1	7.22	0.0087	—
Welch, Hann segments $N_s = N/4$, 50% overlap	13.4	5.84	0.0005	−31 dB

$N = 256$, $R = 8/(N\Delta)$, $2\alpha = 8$, $L = 7$. The half-power width is in units of $1/(N\Delta)$. Leakage is the fraction of kernel mass at $|f| > R$, which for the Gaussian is the tail of the kernel itself. The side lobe is the peak of the kernel beyond $1.5\times$ the half-power half-width, relative to its maximum. For the parabola that point lies on the skirt of the kernel itself, and beyond $1.5R$ the kernel is below −38 dB. The Welch kernel has no flat top and a main lobe reaching $8/(N\Delta)$. Widths are counted on an FFT grid of 4096 points, which is why the ideal box reads 8.06 instead of 8. The figures use finer grids, which changes widths and leakages in the second significant figure. The values are computed as in Appendix B, and the kernels are drawn in Fig. S1.

The spectral window of a short Hann segment has no flat top, and the base of its main lobe, $4/(N_s\Delta)$, is almost three times its half-power width. As a result it separates two equal tones only when they are $10.8/(N\Delta)$ apart, compared with $9.0/(N\Delta)$ for the Hann-tapered box of the same half-power width. Bronez found that multitaper estimation outperforms Welch’s method when both resolution and leakage are matched [26]. The two findings are consistent, because giving the Welch kernel sharp edges requires longer segments, and ν then falls below Thomson’s. The routes can also be combined. Riedel, Sidorenko and Thomson found that a few tapers followed by smoothing performed best on plasma data [23], and we find the same. Two well-concentrated Slepian tapers ($\alpha = 2$) followed by the box give $\nu = 13.3$, a side lobe at −34 dB, and errors of 1.3, 1.9 and 1.6 dB.

3.2 EEG With Seizures

In Fig. 8 the ictal rhythm begins near 7.5 Hz with harmonics up to 25 Hz, slows over about a minute, and ends in post-ictal suppression. This descending pattern is often called a chirp and is typical of an evolving seizure. The second clip contains two such seizures, each with a stack of harmonics that descends from 20 Hz to 3 Hz (not shown). These descriptions are based on reading the spectrograms. The unsmoothed periodogram shows the chirps but is very noisy. The multitaper and smoothed-periodogram spectrograms cannot be told apart by inspection, and the spectra at a pre-ictal and an ictal instant (Fig. 8d,f) show all three practical estimators resolving the same harmonics. Over all 599 windows the two spectrograms differ by a median of 0.9 dB (Fig. 8h), and by 0.7 dB in the second clip. With the Hann taper and the box the medians are 1.5 and 1.2 dB. These differences are smaller than the noise of the estimators, which is 2.5 and 2.1 dB for $\nu = 6$ and $\nu = 8.7$, because the two estimates share most of their noise.

Band powers agree more closely than individual spectral values, because integration over a band averages the noise that distinguishes the estimators (Table 3). In Fig. 8 the ictal rise in

theta, alpha and beta power is 17 to 18 dB by every estimator, and the estimators agree to within 0.7 dB. The disagreement with the multitaper estimate in individual windows is about 1 dB for the Hann route, 0.5 dB for Welch’s method, and 0.1 to 0.3 dB for recipe (b’). Conclusions drawn from band powers in these recordings would not depend on the route.

3.3 Sleep Spindles

Fig. 9 shows the trade between time resolution and frequency resolution on five sleep spindles, each 0.5 to 0.9 s long. With a 1-second window a spindle appears 0.9 s long, and with a 4-second window it appears 3.0 s long. Its width in frequency falls from 2.6 Hz with the 2 Hz kernel to 1.5 Hz with the 1 Hz kernel. The white box in each panel is the resolution of that panel, and its area gives ν through (8). With windows of 2 and 4 s the two routes differ by a median of 0.7 dB over the panel, and the smoothed periodogram has $\nu = 8.7$ where the multitaper estimate has 6. With the 1-second window the time–bandwidth product is $\alpha = 1$. The multitaper estimate then has a single taper, $\nu = 2$ and a kernel 1.1 Hz wide, and the smoothed periodogram has $\nu = 4.7$ and a kernel 2.1 Hz wide. A second Slepian taper would raise ν to 4, but at $\alpha = 1$ it has 25% of its energy outside the band. At the small time–bandwidth products that brief events require, a smoothing kernel can be given any width, and the number of tapers can only be a small integer.

3.4 A Check: Each Route at Its Best Setting

The comparisons above fix the resolution, as a user would. As a check that the conclusions do not depend on that choice, Table 4 gives the smallest error that each family reached when its parameters were tuned to minimize the error over the whole band. These settings are not recommendations. On the EEG-like spectrum with $N = 400$ the best kernel is 4.2 Hz wide at half power, three times the width of the alpha peak, because most of the band has no features. A user who needs to resolve

Table 3: Band Powers From the Estimators Used in Practice on the Two Seizure Clips

clip	band	periodogram	multitaper ictal minus pre-ictal	Hann+box pre-ictal band power	recipe (b')	Welch	periodogram median	Hann+box difference from multitaper	recipe (b') per window	Welch (dB)
Fig. 8 (C4)	delta (1–4 Hz)	+6.6	+6.3	+6.3	+6.4	+6.3	0.64	1.08	0.23	0.61
	theta (4–8 Hz)	+17.8	+17.8	+17.7	+17.6	+17.6	0.41	0.97	0.31	0.64
	alpha (8–13 Hz)	+17.9	+18.0	+17.6	+18.1	+18.1	0.36	0.97	0.26	0.61
	beta (13–30 Hz)	+17.8	+17.7	+17.5	+17.8	+17.8	0.22	0.67	0.16	0.38
second clip (Fp1)	delta (1–4 Hz)	+8.0	+8.1	+7.8	+8.3	+8.2	0.46	0.95	0.28	0.73
	theta (4–8 Hz)	+7.8	+7.7	+7.8	+7.8	+7.7	0.31	0.78	0.19	0.56
	alpha (8–13 Hz)	+4.3	+4.4	+4.2	+4.5	+4.5	0.21	0.58	0.12	0.44
	beta (13–30 Hz)	+5.6	+5.6	+5.4	+5.5	+5.5	0.13	0.41	0.11	0.38

One channel of each clip, 599 windows of 2 s per clip (75 ictal and 120 pre-ictal windows for the clip of Fig. 8; 279 and 43 for the second clip, which contains two seizures and is not shown). Left: the mean ictal band power minus the mean pre-ictal band power (the windows in the two minutes before the first seizure, of which the second clip has only 43), in dB, by each estimator. Right: the median over all windows of the absolute difference between each estimator’s band power and the multitaper estimate’s, in dB. The periodogram is that of the cosine-tapered window. Resolution $R = 2$ Hz throughout ($\alpha = 2$; box of width 2 Hz after a Hann taper; recipe (b’), a parabola of half-power width 2 Hz after a 25% cosine taper; 0.7-second Welch segments). The degrees of freedom are $\nu = 6.0$ for the multitaper estimate, 4.8 for the Hann route, 8.7 for recipe (b’) and 7.7 for Welch’s method.

the peak would choose a narrower kernel and accept more noise.

With that caveat, the families agree. In each of the four cases the tapered families lie within 12% of one another, and a single taper followed by smoothing is never behind the multitaper estimate. Multitaper estimates with Slepian tapers and equal weights are 3 to 8% behind the single taper followed by smoothing, and adaptively weighted ones are 4 to 11% behind. The adaptive weights protect a default setting against leakage. A setting that has been tuned does not need them.

The best single taper was a cosine taper of 10% or 25% in every case. The Hann window was 17 to 24% behind it, and the parabola was 2 to 4% ahead of the box (`demos/taper_and_kernel.py`). The 25% cosine taper was within 3% of the best taper in every case, including records of 128 samples of the AR(4) process, for which the 10% taper leaks. Without any taper, the smoothed periodogram is the most accurate estimator on the EEG-like spectrum and the least accurate, by 8 to 11 dB, on the AR(4) process.

The kernel of (24) was tuned in the same way, over the taper and over the multitaper estimate to which it is matched. Matched without a taper to the best multitaper estimate, it reproduces the error of that estimate: 1.960 and 1.395 dB on the EEG-like spectrum, against 1.962 and 1.399 dB (`demos/tuned.precision.py`). Its own best settings are matched to a single Slepian taper with a large time–bandwidth product, $\alpha = 16$ to 160, whose kernel is a smooth bell that widens with α . At its best the matched kernel is therefore another smoothing kernel with one parameter. Searched over widths as finely, the parabola is as accurate: its best errors are 2.414, 1.363, 1.836 and 1.327 dB in the four cases, against 2.425, 1.363, 1.838 and 1.328 dB (`demos/taper_and_kernel.py`). On the AR(4) process the matched kernel requires a taper. Without one, 8.9% or more of its estimates are negative at every setting. A kernel matched to the multitaper estimate is therefore the right choice for reproducing that estimate, and not for improving on it. The parabola (9) is as accurate, has one parameter, and cannot

give a negative estimate.

The lower rows of the table give the estimators at the bandwidth used in the figures. At that bandwidth recipe (b’) has a smaller error than the adaptively weighted multitaper estimate in all four cases.

The settings above use one width at all frequencies. The best width (10) varies with the curvature of the spectrum, so we also chose the width separately at each frequency, on one set of records, and scored the choice on a second set. With a 25% cosine taper and a parabola the error falls from 2.42, 1.41, 1.94 and 1.41 dB in the four cases to 1.69, 1.01, 1.39 and 1.00 dB, a gain of 28 to 30%. The multitaper estimate with equal weights falls to 1.77, 1.15, 1.42 and 1.08 dB, and with adaptive weights to 1.87, 1.26, 1.47 and 1.17 dB. The gain from choosing the width well is several times the difference between the routes. The choice uses the true spectrum, so these values are bounds and not the results of a procedure. A width that varies with frequency is simple to apply to a periodogram. A multitaper estimate requires a different set of tapers for every width (`demos/local_width.py`).

4 Discussion

4.1 Recipes for the Three Routes

The three routes are close enough in accuracy that the choice among them can be made on practical grounds. In the boxes below, N is the number of samples, Δ the sampling interval and R the desired resolution in hertz, so that $\alpha = N\Delta R/2$ by (3). We recommend setting R to about the width of the narrowest feature that must be preserved (Section 2.4.2), and the window length to the duration of the briefest event that must be resolved in time. The degrees of freedom then follow from (8).

Exact versions. These give the same estimate, to machine precision for (a) and (b) and to a fraction of a decibel

Table 4: Error of Each Family at Its Best Setting

Family	AR(4) process		EEG-like spectrum	
	$N = 256$	$N = 1024$	$N = 400$	$N = 1024$
One taper, then smooth	2.42	1.37	1.89	1.36
A few Slepian tapers, then smooth	2.46	1.49	1.87	1.37
Welch, Hann segments	2.48	1.39	1.90	1.36
Multitaper, sine tapers, equal weights	2.47	1.41	1.99	1.41
Multitaper, Slepian tapers, equal weights	2.61	1.45	1.96	1.40
Multitaper, Slepian tapers, adaptive weights	2.69	1.48	2.00	1.41
One taper, then kernel matched to a multitaper estimate	2.43	1.36	1.84	1.33
No taper, then smooth	13.98	9.71	1.84	1.33
<i>At the fixed resolution $R = 8/(N\Delta)$</i>				
Cosine taper (25%), then parabola, recipe (b')	2.87	1.59	1.95	1.60
Hann periodogram, then box	2.98	2.18	2.45	2.18
Multitaper, $L = 7$, adaptive weights	2.99	2.39	2.05	1.78
Multitaper, $L = 7$, equal weights	8.67	5.10	2.03	1.76

Root-mean-square error in dB over 1000 simulated records (500 for adaptive weights) and about 200 frequencies. The best settings, in the order of the columns, were: one taper, a 25% cosine taper with a parabola of half-width $3.5/(N\Delta)$, then 10% cosine tapers with parabolas of half-width $8/(N\Delta)$, $6/(N\Delta)$ and $10/(N\Delta)$; Slepian tapers with equal weights, $(\alpha, L) = (4.5, 5)$, $(8.5, 13)$, $(4.5, 9)$ and $(7.5, 15)$; adaptive weights, $(5, 5)$, $(9, 13)$, $(5, 9)$ and $(8, 15)$; sine tapers, $L = 5, 14, 9$ and 15 ; Welch, 18 segments of 84 samples, 57 of 128, 33 of 80 and 57 of 128, all at 7/8 overlap; matched kernel (24), a 25% cosine taper matched to $(16, 1)$, a 10% cosine taper matched to $(112, 1)$, and no taper matched to $(64, 1)$ and $(160, 1)$, searched over α from 1 to 32 with every L and from 40 to $N/4$ with $L \leq 4$. Settings of the matched kernel at which 0.1% or more of the estimates were negative were excluded, and its error is computed over its positive estimates. Monte Carlo precision: the bootstrap standard errors of the entries are 0.004 to 0.013 dB, and up to 0.10 dB for the untapered and $L = 7$ equal-weight rows. The rows share records, and the paired-bootstrap 95% interval of each tapered row relative to the first row of its block is within $\pm 1\%$. Rescored on 2000 new records, no tapered entry changes by more than 0.02 dB and no ranking within either block changes. Computed by `demos/tuned.comparison.py`, `demos/taper.and.kernel.py` and `demos/tuned.precision.py`.

for (c). The windows in (b) and (c) follow from requiring equality with (a) (identity (16), Appendix C).

- (a) *Multitaper*. Compute all N Slepian tapers for (N, α) and their concentrations λ_i . Average the eigenspectra with weights $\lambda_i/2\alpha$.
- (b) *Smooth*. Compute the periodogram of the untapered record, zero-padded to at least $2N$ points. Convolve it along frequency with a box of width R .
- (c) *Average*. Use a sinc window $h_k = R\Delta \text{sinc}(R\Delta k)$ many times longer than the record. Place it at every integer offset n , including offsets at which it extends beyond the zero-padded ends. Sum the periodograms $\Delta |\sum_k h_{k-n} x_k e^{-i2\pi k f \Delta}|^2$ over n and divide by 2α .

The estimate that these share contains the leakage of the untapered periodogram.

Versions for practical use. These remove the leakage in three different ways, so they are close but not identical.

- (a') *Multitaper*. Keep the $L = 2\alpha - 1$ tapers with $\lambda_i \approx 1$ and weight them adaptively, or keep $L = 2\alpha - 2$ with equal weights. Expect $\nu \approx 2L$ where the spectrum is high and about half that where it is far below its maximum.
- (b') *Smooth*. Taper the first and last eighth of the record

with half a cosine (a 25% cosine, or Tukey, taper) and scale the taper to unit energy. Compute the squared FFT, zero-padded to at least $2N$ points, or more for a finer display grid. Convolve with a parabola of half-width $R/\sqrt{2}$, whose half-power width is R . When each kernel is used at its best width, the parabola has the smallest asymptotic mean square error of any non-negative kernel (Section 2.4.3). At the fixed half-power width used here it has 20% more bias and 15% less variance than the box. Expect $\nu \approx 2 \cdot 2\alpha$, a kernel without a flat top whose base is 1.4 times its half-power width, and skirts whose level beyond $1.5R$ depends on the time-bandwidth product. For any N they are below -25 , -31 and -38 dB for $2\alpha = 4, 6$ and 8 . A box of width R has the same half-power width and a flat top, and was 2 to 4% less accurate in Table 4. If the dynamic range is much larger than 65 dB, use a Hann window. Its skirts are at least 25 dB lower, and it halves ν . The same applies near a spectral line such as mains interference. A line 30 dB above the background in the estimate biases (b') by 1 dB or more up to $2.3R$ from the line for $2\alpha = 4$ and $1.4R$ for $2\alpha = 8$, and the multitaper estimate with $L = 2\alpha - 1$ up to $3.1R$ and $2.5R$ (`demos/line.leakage.py`). Within about $2.5R$ of such a line, remove the line first or use a Hann window, which keeps the bias within R .

- (c') *Average*. Use Hann-tapered segments of duration $N_s \Delta \approx 1.4/R$ that overlap by half, and average their

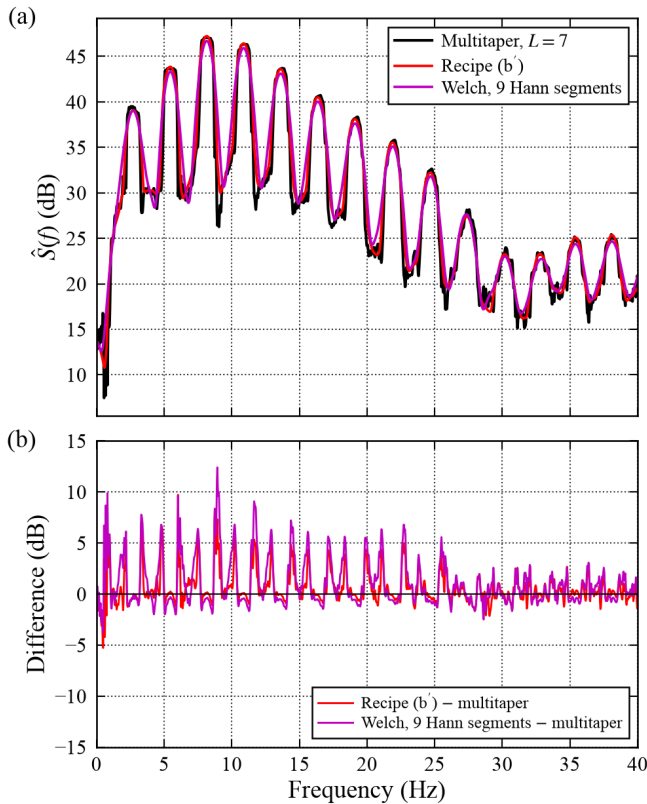


Figure 7: The estimators used in practice agree to within their own noise. (a) The EEG epoch of Fig. 1 estimated by Thomson’s method with $L = 7$ tapers ($\nu = 14$, black), by a cosine-tapered periodogram smoothed with a parabola of half-power width R , recipe (b’) ($\nu = 16.7$, red), and by Welch’s method with nine Hann segments of 1.5 s stepped by 0.81 s ($\nu = 17.6$, magenta). Their half-power widths are 0.91, 1.03 and 0.97 Hz. (b) Differences from the multitaper estimate: median absolute difference 0.6 dB (smoothed) and 0.9 dB (averaged), 95th percentiles 4.2 and 5.9 dB. The medians are smaller than the estimators’ own noise ($4.3\sqrt{2/\nu}$ gives 1.6, 1.5 and 1.5 dB). The largest differences are in the valleys between harmonics, which the side lobes of the seventh Slepian taper fill in.

periodograms. Expect ν of about twice the number of segments, a kernel with no flat top whose main lobe has a base of $4/(N_s\Delta)$, and leakage set by the Hann window of N_s samples.

We suggest route (a’) when the dynamic range is unknown and may be large, when Thomson’s F-test for spectral lines is needed, or when coherence and cross-spectra are required, because the statistical properties of multitaper cross-spectral estimates are well established [27]. Route (b’) is preferable when the bandwidth should vary with frequency, which is difficult to arrange with tapers or segments, and for time–frequency displays, where smoothing along both axes is natural. It is

also the easiest of the three to interpret. Route (c’) suits long records and pipelines that already compute it. At matched half-power width it has the most degrees of freedom and the poorest two-tone resolution. For any route, ν determines the error bar. The 95% confidence interval of Section 2.2 extends from -3.3 to $+5.3$ dB around the estimate for $\nu = 8.9$, from -2.7 to $+4.0$ dB for $\nu = 14$, and from -2.5 to $+3.6$ dB for $\nu = 16.7$.

4.2 Relation to Prior Work

Established results The results assembled here are established, but they are scattered across sources that most users of spectral estimation do not read. Identity (16) is Thomson’s equation (8.3) [6], which takes up half a page of a forty-page paper. The fact that any quadratic estimator is a weighted multitaper estimator, with the eigenvectors of its matrix as tapers, appears in the textbook of Percival and Walden [11]. Riedel and Sidorenko developed it for kernel smoothers [18] and Walden for multivariate estimators [27]. Riedel and Sidorenko also showed that the equivalent tapers of a kernel smoother resemble sine tapers, and noted that a sine-taper estimate has low bias because the side lobes of neighbouring Fourier coefficients cancel. The degrees-of-freedom formula (15) is Satterthwaite’s approximation [19] applied to this decomposition. Welch showed that averaging the periodograms of a sliding window smooths the periodogram [2], and Nuttall and Carter gave the general form with combined time and lag weighting [10]. Hansson-Sandsten approximated the matrix of the Thomson estimator by that of a Welch estimator, in the mean-square sense, for real-time use [22]. Thomson and Chave remarked that section averaging and multiple-window estimates are mathematically similar [24]. McCloud, Scharf and Mullis showed that lag-windowed estimators have multiple-window implementations and concluded that the two are roughly equivalent for smooth spectra [21], and the textbook of Stoica and Moses presents the Slepian estimator as an approximation to the smoothed periodogram [20]. Abreu and Romero proved that the L -taper kernel converges to a box and gave the rate [28]. Karnik, Romberg and Davenport used (16) in matrix form in a fast multitaper algorithm [25]. Riedel, Sidorenko and Thomson compared multitaper estimates with smoothed tapered periodograms [23], and Bronez compared them with Welch’s method [26]. Harris catalogued the properties of windows [29]. Most of this literature is written in terms of eigenvalue problems, error rates and matrix algorithms. The tutorials, software and textbooks from which neuroscientists learn spectral estimation [3, 7, 8, 30, 31] do not relate the three methods to one another.

The view that smoothing is obsolete Thomson and Chave wrote that periodograms and smoothed periodograms are “hopelessly obsolete” [24], and many users regard multitaper estimation as a different and better method. The statement is correct for a periodogram without a taper. By (16) that estimate is a multitaper estimate with all N tapers, and Thomson observed that the poorly concentrated ones bias it [6].

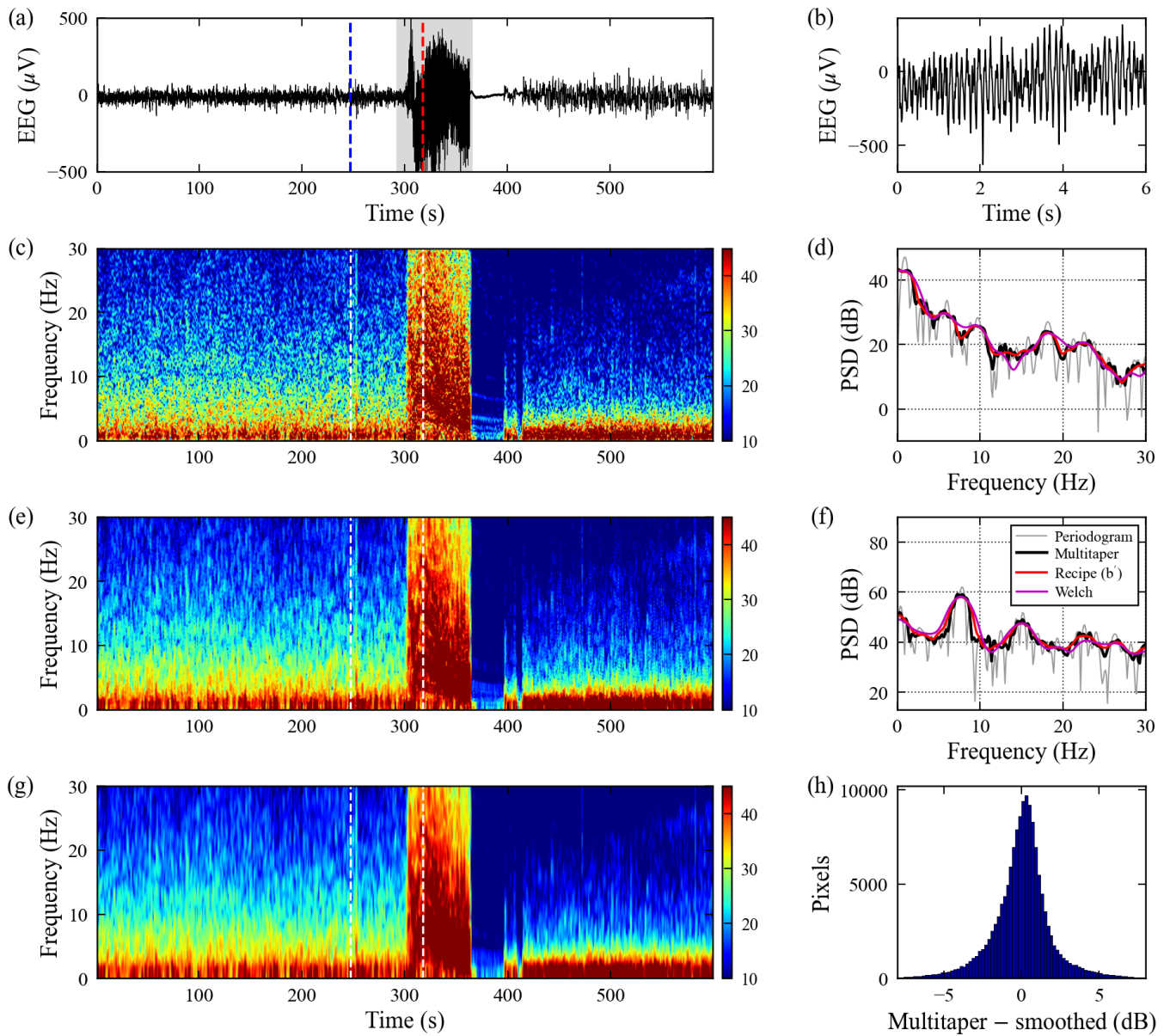


Figure 8: Three estimators of a seizure. (a) Ten minutes of de-identified scalp EEG at 200 Hz (ELROND seizure-annotation set), channel C4 against the common average, displayed between $\pm 500 \mu V$; the seizure is shaded and the dashed lines mark the instants used in (d) and (f). (b) Six seconds of the ictal rhythm from $t = 318$ s. (c, e, g) Spectrograms from 599 windows of 2 s stepped by 1 s: the periodogram after a 25% cosine taper ($\nu = 2$), the multitaper estimate with $\alpha = 2$, $L = 3$ ($\nu = 6$), and the same periodogram smoothed with a parabola 2 Hz wide at half power ($\nu = 8.7$), on one colour scale of 10 to 45 dB. (d, f) Spectra at the pre-ictal and ictal instants by all three, plus Welch’s method with 0.7-second Hann segments ($\nu = 7.7$). (h) Distribution over all pixels of the difference between (e) and (g). The channel is the one with the largest ictal rise in 2 to 20 Hz power.

In Table 1 it errs by 13.7 dB where the spectrum is low. The statement does not hold for a tapered periodogram. The Hann-tapered, smoothed periodogram errs by 1.4 dB in the same region, and the adaptively weighted multitaper estimate by 1.7 dB. With a Hann taper the smoothed periodogram gives up efficiency instead. At the same design bandwidth it has $\nu = 8.9$ and the multitaper estimate has 14. The loss belongs to the Hann window and not to smoothing. With a 25% cosine taper the smoothed periodogram has $\nu = 14.8$ with the box

and 16.7 with the parabola, and when every route is tuned it is at least as accurate as the multitaper estimate in each of our test cases (Table 4). Riedel, Sidorenko and Thomson likewise found a smoothed estimate with a few tapers to be the most accurate of those they compared [23]. Multitaper estimation is one member of a family of estimators of similar accuracy. Its particular strengths are a bound on the leakage of each taper that holds for any spectrum, the F-test for spectral lines, and an established theory for cross-spectra.

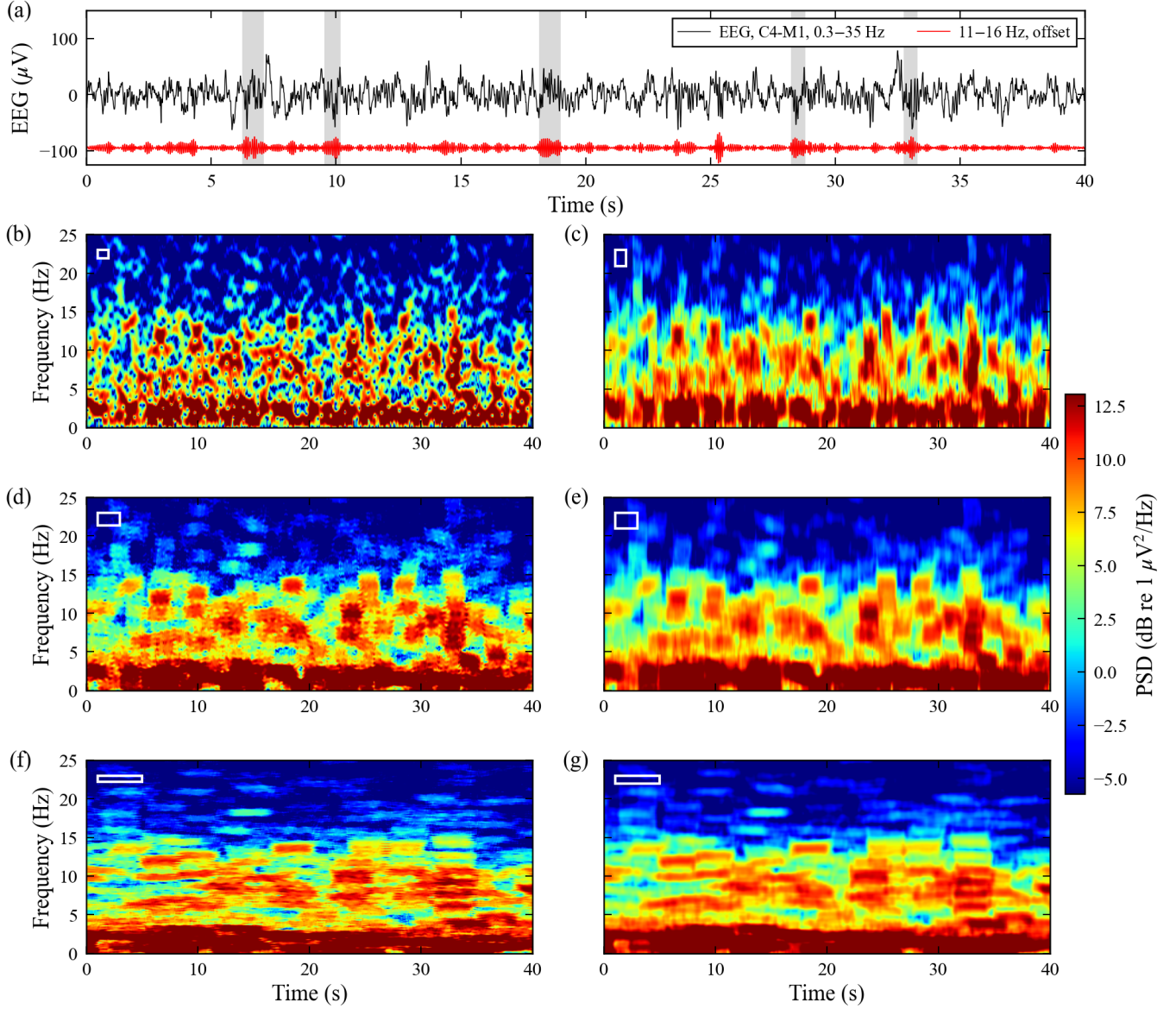


Figure 9: Sleep spindles at three resolutions. (a) Forty seconds of stage N2 sleep, channel C4-M1, filtered between 0.3 and 35 Hz and between 11 and 16 Hz (offset), with five spindles shaded. (b–g) Spectrograms from windows of 1 s (b, c), 2 s (d, e) and 4 s (f, g) stepped by 0.1 s, by the multitaper estimate with $L = 2\alpha - 1$ tapers, at least one (left), and by a cosine-tapered periodogram smoothed with a parabola (right), on one colour scale. The half-power widths of the kernels are 1.1 and 2.1 Hz for the 1-s windows ($\nu = 2.0$ and 4.7), 1.7 and 2.0 Hz for the 2-s windows ($\nu = 6.0$ and 8.7), and 0.8 and 1.0 Hz for the 4-s windows ($\nu = 6.0$ and 8.7). The white box is the window length by the half-power width, so it shows the resolution of the panel, and twice its area is approximately ν . Short windows keep the spindles brief and blur them in frequency, and long windows do the opposite.

Contributions of this paper We combine these results in a single description, the bank of tapers, and work out its practical consequences. The split (21) turns the approximate equivalence of smoothing and multitaper estimation into an exact one, by writing out the remainder as a cross term and measuring it. The sliding-window form of the identity relates the averaging route exactly to the other two, and identifies the Slepian sequences as the principal components of a sliding sinc window. The eigen-weight ladder explains a standard

degrees-of-freedom formula in terms that apply to all three routes, since the variance cost of a single taper corresponds to the slope of its ladder. Tables 1 and 2 list the cost of each route and the conditions under which each performs best, including the adaptively weighted estimate. The bandwidth rule and the recipes allow the parameters to be chosen from the features of the data. Bruns showed in a similar way that Fourier, Hilbert and wavelet analyses of neural signals are formally equivalent [32]. The bank-of-tapers description covers those

methods as well. A Morlet scalogram, for example, is a single-taper periodogram whose Gaussian taper shortens as frequency increases.

4.3 Limitations

The identity is exact only for the eigenvalue-weighted estimate with all N tapers, and its sliding-window form only in the limit of a long window. The versions used in practice are close but not equal. A smoothing kernel matched to the multitaper kernel removes the difference in expected value and not the difference in individual estimates.

The comparison in Table 1 uses a single process, a fourth-order autoregression with two peaks and a range of 65 dB. The estimators are compared at matched design bandwidth and not at matched ν . The median error over the whole band is dominated by the region of low power, where leakage is largest. For this reason the table also reports the error at the peaks, where the ranking of the estimators is reversed. Table 4 adds a second spectrum and tunes every route, but the tuning uses the true spectrum, which is not available in practice, and it scores one measure of error over the whole band. Other spectra, spectral lines and very short records will rank the routes differently. The adaptive weights are those of Thomson's original iteration. We did not use the jackknife error estimates [24] or the quadratic-inverse estimates [33] that were developed later.

The kernels, degrees of freedom and leakage values describe the estimators for a stationary Gaussian process and were computed on a finite FFT grid. The spectrograms assume stationarity within each 2-second window, an assumption that fails at the onset of a seizure. The EEG evidence consists of two clips with one channel each, analyzed with equal-weight multitaper estimates. We did not consider coherence or cross-spectra, inference beyond the χ^2_ν interval, or data-driven bandwidth selection.

5 Conclusion

Averaging periodograms, smoothing a periodogram and multitaper estimation are the same estimator, a bank of tapers with weights that can be built from one window in three ways. When all Slepian tapers are used with their eigenvalues as weights, the multitaper estimate equals a smoothed periodogram exactly, and in the limit of a long window it equals an average of sliding-window periodograms. A multitaper estimate smooths the Fourier transform before squaring it, and a smoothed periodogram smooths after. The estimator used in practice is this estimate with its leaky components removed. This description expresses the multitaper parameters in terms that users of the other two methods already know. The taper controls the skirts of the kernel, and the box width or segment length controls the width of its main lobe. For every route the degrees of freedom are about twice the product of the window length and the frequency resolution, so the user chooses the resolution in time and in frequency and the noise follows. A

lightly tapered and smoothed periodogram controls leakage as well as an adaptively weighted multitaper estimate, and is as accurate when both are tuned. A heavy taper such as the Hann window halves the degrees of freedom. Among non-negative smoothing kernels, each at its best width, the parabola has the smallest asymptotic mean square error, and for white noise no quadratic estimator with a given kernel has a smaller variance than the untapered smoothed periodogram. Without a taper, a smoothed or averaged periodogram is biased by leakage whenever the spectrum is low. For most EEG applications any of the three routes is adequate, and the choice can be made for convenience.

Data and Code Availability

The code is public at <https://github.com/bdsp-core/spectral-smoothing-vs-multitaper>. The scripts in the repository generate every figure and table. The identities (16) and (17), the sliding-window convergence at $N_s = N$ to $32N$, the formulas of Appendix B and their Monte Carlo checks are implemented as tests. The simulated data and the 128 Hz EEG epoch of Figs. 1 and 7 are included. The EEG shown in Figs. 8 and 9 and used in Table 3 is included as de-identified excerpts, one channel of each recording. The complete recordings are clinical recordings from the Brain Data Science Platform and are available to credentialed users under its data-use agreement. The scripts print every number quoted from the clips, and the printed output is in the repository.

A Proof of the Identity

Let Φ_R be the matrix (11). The Slepian tapers are its eigenvectors, $\Phi_R h^{(i)} = \lambda_i h^{(i)}$, $\|h^{(i)}\| = 1$ [9], so $\Phi_R = \sum_{i=1}^N \lambda_i h^{(i)} h^{(i)'}$ and $\sum_i \lambda_i = \text{tr } \Phi_R = NR\Delta = 2\alpha$. By (12), $\hat{S}^{(i)}(f) = \Delta |h^{(i)'} x_f|^2$, and

$$\begin{aligned} \sum_{i=1}^N \lambda_i \hat{S}^{(i)}(f) &= \Delta x_f^* \left(\sum_i \lambda_i h^{(i)} h^{(i)'} \right) x_f \\ &= \Delta \sum_{k,l} (\Phi_R)_{k,l} x_k x_l e^{-i2\pi(k-l)f\Delta} \\ &= \Delta \sum_{\tau=-(N-1)}^{N-1} \frac{\sin(\pi R \Delta \tau)}{\pi \tau} \hat{s}_\tau e^{-i2\pi \tau f \Delta}, \quad (25) \end{aligned}$$

with $\hat{s}_\tau := \sum_k x_k x_{k+\tau}$, collecting terms by lag; the factor at $\tau = 0$ is $R\Delta$, and the sign of the exponents does not matter because both sums are symmetric under $k \leftrightarrow l$. By (4), $\hat{S}^p(f') = (\Delta/N) \sum_\tau \hat{s}_\tau e^{-i2\pi \tau f' \Delta}$, and $\int_{f-R/2}^{f+R/2} e^{-i2\pi \tau f' \Delta} df' = e^{-i2\pi \tau f \Delta} \sin(\pi R \Delta \tau) / (\pi \tau \Delta)$, so the sum equals $N\Delta \int_{f-R/2}^{f+R/2} \hat{S}^p(f') df'$. Divide by $2\alpha = N\Delta R$. $\square$

B Kernels, Banks and Variance of Quadratic Estimators

Every estimator in this paper is $\hat{S}(f) = \Delta x_f^* Q x_f$ with Q real symmetric positive semidefinite and independent of f [11, 18]: $Q = \sum_i c_i h^{(i)} h^{(i)*}$ for a multitaper estimate; $Q_{k,l} = h_k g_{k-l} h_l$ for a periodogram tapered with h and smoothed with the lag window g , whose transform $G(f) = \Delta \sum_\tau g_\tau e^{-i2\pi\tau f\Delta}$ is the kernel; $Q_{k,l} = \sum_n h_{k-n} h_{l-n} / \sum_n \sum_k h_{k-n}^2$ for averaged periodograms of a window h placed at offsets n .

Bank. $Q = \sum_i c_i h^{(i)} h^{(i)*}$ with orthonormal $h^{(i)}$ and $c_i \geq 0$, so every Q is a multitaper estimate (14).

Kernel. $\mathbf{E}\{\hat{S}(f)\} = \int \mathcal{K}(f - f') S(f') df'$ with $\mathcal{K}(f) = \Delta \sum_\tau q_\tau e^{-i2\pi\tau f\Delta}$ the transform of the lag sums $q_\tau = \sum_k Q_{k,k+\tau}$; equivalently $\mathcal{K} = \sum_i c_i |H^{(i)}|^2 / \Delta$. For a tapered, smoothed periodogram $\mathcal{K} = G * |H|^2 / \Delta$; for averaged periodograms $\mathcal{K} = |H|^2 / \Delta$ for the segment window normalized to unit energy, whatever the overlap; for $Q = \Phi_R$, $\mathcal{K}$ is $N\Delta$ times the indicator of $[-R/2, R/2]$ convolved with $N\Delta \mathcal{D}_N^2$, which is why two rows of Table 2 coincide.

Degrees of freedom. For white Gaussian noise of unit variance, $x_f^* Q x_f$ has mean $\text{tr } Q_f$ and variance $2 \text{tr } Q_f^2$, with $(Q_f)_{k,l} := Q_{k,l} \cos(2\pi(k-l)f\Delta)$ the real part of Q conjugated by the modulation, so $\nu = (\text{tr } Q_f)^2 / \text{tr } Q_f^2$, the chi-square approximation of Satterthwaite [19]. At an interior frequency each modulated taper is orthogonal to its conjugate, each term of the bank is an independent χ_2^2 , and $\nu = 2(\sum_i c_i)^2 / \sum_i c_i^2$, equation (15). The two formulas agree to within 3% for the three estimators tested (multitaper, Hann-then-box and Welch); Table 2 uses the first, and Monte Carlo on white noise (3000 draws) agrees with it to within 15% for the same three (repository tests).

Resolution and leakage. Resolution is the half-power full width of H ; leakage is the fraction of H outside a stated band; the peak side lobe is the maximum of H beyond $1.5\times$ the half-power half-width. A practitioner's convolution of a zero-padded periodogram with a kernel on the same grid is exactly the lag-window estimate above once the padding reaches $2N$ points, because the periodogram's lag sequence vanishes beyond $|\tau| = N - 1$.

C Averaging Over a Sliding Window Is Smoothing

C.1 Continuous Time: the Limit of All Overlaps

Let $x(t)$ be the record, zero outside $[0, T]$, with transform $X(f) := \int x(t) e^{-i2\pi f t} dt$, and $h(t)$ a real symmetric window with transform $H(f)$. Windowing at shift τ gives $x(t)h(t-\tau)$, with transform

$$Y_\tau(f) = \int X(f') H(f - f') e^{-i2\pi(f-f')\tau} df'.$$

Welch's method with shifts $n\delta$ averages $|Y_{n\delta}(f)|^2$ over n . Let the overlap grow without bound, $\delta \rightarrow 0$: the average becomes

a Riemann sum for an integral over the shift, which can be evaluated in closed form,

$$\begin{aligned} & \int_{-\infty}^{\infty} |Y_\tau(f)|^2 d\tau \\ &= \iint X(f') X^*(f'') H(f - f') H^*(f - f'') \\ & \quad \times \left[\int_{-\infty}^{\infty} e^{i2\pi(f' - f'')\tau} d\tau \right] df' df'' \\ &= \int |X(f')|^2 |H(f - f')|^2 df' \\ &= (|H|^2 * |X|^2)(f), \end{aligned} \tag{26}$$

because the bracket is $\delta(f' - f'')$. The sum of the periodograms of all shifted windows is the raw periodogram smoothed with the window's own kernel $|H|^2$; dividing by $T \int h^2$, the trace of the quadratic form, makes the estimate unbiased for white noise. The shift integral must run over every τ at which the window touches the record, including shifts at which it extends beyond the ends. These shifts are needed to produce the delta function, and restricting τ to $[0, T]$ leaves an edge term that vanishes only when T is much longer than the window. For a Gaussian window of standard deviation σ_t the kernel is Gaussian with standard deviation $1/(2\sqrt{2}\pi\sigma_t)$, so sliding a Gaussian window is equivalent to smoothing with a Gaussian. This calculation, with a Gaussian window, was the content of an unpublished 2001 note by one of us, written before the connection to multitaper estimation was known; the result is Welch's [2] and is contained in Nuttall and Carter's framework [10].

C.2 Discrete Time and a Finite Record

In discrete time the shift is an integer, and the sum over all shifts gives the result exactly, with no limit required. Let h have N_s samples, h_k for $k = 0, \dots, N_s - 1$, and place it at offset n , so that it covers the samples $n, \dots, n + N_s - 1$ of the record, discarding the part outside the record. Summing over offsets gives $Q_{k,l} = \sum_n h_{k-n} h_{l-n}$. If the offsets run over every integer for which the window touches the record, $n = -(N_s - 1), \dots, N - 1$, the sum equals the autocorrelation $r_h(k - l)$ for every pair inside the record, so Q is Toeplitz and the estimator is the lag-window estimate of the untapered periodogram with lag window r_h and kernel $|H(f)|^2 / \Delta$. This is exact for a finite record; with a larger step or without overlap it holds approximately in the interior. The equivalence is for the *untapered* periodogram: a tapered periodogram that is then smoothed has $Q_{k,l} = h_k g_{k-l} h_l$, which is not Toeplitz and is not the average of any single sliding window, though it is a time-and-lag weighting in the sense of [10]. For the sinc window, $r_h(\tau) \rightarrow R\Delta \text{sinc}(R\Delta\tau)$ as $N_s \rightarrow \infty$, the lag sequence of Φ_R ; the relative error in Q falls as $1/N_s$ because the truncated sinc's autocorrelation differs from the sinc by a term of that order. Hence (17), and the Slepian sequences are the principal components of the shifted sinc windows.

D The Kernel of Least Mean Square Error

Let the spectrum be smooth near f with relative curvature $\kappa := |S''(f)/S(f)|$, and let G be a non-negative kernel of unit area that is much wider than $1/(N\Delta)$. To leading order the relative mean square error of the smoothed periodogram is

$$J[G] = \frac{\kappa^2}{4} \left(\int u^2 G du \right)^2 + \frac{C_h}{N\Delta} \int G^2 du. \quad (27)$$

The first term is the squared bias and the second is the variance. We minimize J subject to $\int G du = 1$ and $G \geq 0$. With a multiplier η for the first constraint, the first variation vanishes wherever $G > 0$ if

$$\frac{\kappa^2}{2} \mu_2 u^2 + \frac{2C_h}{N\Delta} G(u) - \eta = 0, \quad \mu_2 := \int u^2 G du. \quad (28)$$

The minimizer is therefore a parabola, $G(u) = a(b^2 - u^2)$ for $|u| \leq b$ and zero elsewhere, with $a = N\Delta\kappa^2\mu_2/(4C_h)$. Unit area requires $a = 3/(4b^3)$, and the parabola has $\mu_2 = b^2/5$. Together these give $b^5 = 15C_h/(N\Delta\kappa^2)$, which is (10). Both terms of J are convex in G , so the stationary point is the minimum. One calculation gives the shape and the width. The shape does not depend on the spectrum, and the width depends on it only through the curvature κ .

Acknowledgment

AI tools were used to assist with software development, preparation of the figures and drafting of the manuscript. The authors reviewed and revised all content and take responsibility for it.

References

- [1] M. S. Bartlett, "Smoothing periodograms from time-series with continuous spectra," *Nature*, vol. 161, no. 4096, pp. 686–687, 1948.
- [2] P. D. Welch, "The use of fast Fourier transform for the estimation of power spectra: A method based on time averaging over short, modified periodograms," *IEEE Trans. Audio Electroacoust.*, vol. 15, no. 2, pp. 70–73, 1967.
- [3] S. M. Kay, *Modern Spectral Estimation: Theory and Application*. Englewood Cliffs, NJ: Prentice-Hall, 1988.
- [4] P. J. Daniell, discussion of "On the theoretical specification and sampling properties of autocorrelated time-series," *Suppl. J. Roy. Statist. Soc.*, vol. 8, no. 1, pp. 88–90, 1946.
- [5] R. B. Blackman and J. W. Tukey, *The Measurement of Power Spectra from the Point of View of Communications Engineering*. New York: Dover, 1958.
- [6] D. J. Thomson, "Spectrum estimation and harmonic analysis," *Proc. IEEE*, vol. 70, no. 9, pp. 1055–1096, 1982. See Sec. VIII, eq. (8.3).
- [7] B. Babadi and E. N. Brown, "A review of multitaper spectral analysis," *IEEE Trans. Biomed. Eng.*, vol. 61, no. 5, pp. 1555–1564, 2014.
- [8] M. J. Prerau, R. E. Brown, M. T. Bianchi, J. M. Ellenbogen, and P. L. Purdon, "Sleep neurophysiological dynamics through the lens of multitaper spectral analysis," *Physiology*, vol. 32, no. 1, pp. 60–92, 2017.
- [9] D. Slepian, "Prolate spheroidal wave functions, Fourier analysis, and uncertainty. V: The discrete case," *Bell Syst. Tech. J.*, vol. 57, no. 5, pp. 1371–1430, 1978.
- [10] A. H. Nuttall and G. C. Carter, "Spectral estimation using combined time and lag weighting," *Proc. IEEE*, vol. 70, no. 9, pp. 1115–1125, 1982.
- [11] D. B. Percival and A. T. Walden, *Spectral Analysis for Physical Applications: Multitaper and Conventional Univariate Techniques*. Cambridge Univ. Press, 1993 (2nd ed., *Spectral Analysis for Univariate Time Series*, 2020).
- [12] G. Wahba, "Automatic smoothing of the log periodogram," *J. Amer. Statist. Assoc.*, vol. 75, no. 369, pp. 122–132, 1980.
- [13] C. M. Hurvich, "Data-driven choice of a spectrum estimate: extending the applicability of cross-validation methods," *J. Amer. Statist. Assoc.*, vol. 80, no. 392, pp. 933–940, 1985.
- [14] C. L. Haley and M. Anitescu, "Optimal bandwidth for multitaper spectrum estimation," *IEEE Signal Process. Lett.*, vol. 24, no. 11, pp. 1696–1700, 2017.
- [15] M. B. Priestley, "Basic considerations in the estimation of spectra," *Technometrics*, vol. 4, no. 4, pp. 551–564, 1962.
- [16] V. A. Epanechnikov, "Non-parametric estimation of a multivariate probability density," *Theory Probab. Appl.*, vol. 14, no. 1, pp. 153–158, 1969.
- [17] D. W. K. Andrews, "Heteroskedasticity and autocorrelation consistent covariance matrix estimation," *Econometrica*, vol. 59, no. 3, pp. 817–858, 1991.
- [18] K. S. Riedel and A. Sidorenko, "Minimum bias multiple taper spectral estimation," *IEEE Trans. Signal Process.*, vol. 43, no. 1, pp. 188–195, 1995.
- [19] F. E. Satterthwaite, "An approximate distribution of estimates of variance components," *Biometrics Bull.*, vol. 2, no. 6, pp. 110–114, 1946.
- [20] P. Stoica and R. L. Moses, *Spectral Analysis of Signals*. Upper Saddle River, NJ: Prentice Hall, 2005.

- [21] M. L. McCloud, L. L. Scharf, and C. T. Mullis, "Lag-windowing and multiple-data-windowing are roughly equivalent for smooth spectrum estimation," *IEEE Trans. Signal Process.*, vol. 47, no. 3, pp. 839–843, 1999.
- [22] M. Hansson-Sandsten, "A Welch method approximation of the Thomson multitaper spectrum estimator," in *Proc. 20th Eur. Signal Process. Conf. (EUSIPCO)*, Bucharest, Romania, 2012, pp. 440–444.
- [23] K. S. Riedel, A. Sidorenko, and D. J. Thomson, "Spectral estimation of plasma fluctuations. I. Comparison of methods," *Phys. Plasmas*, vol. 1, no. 3, pp. 485–500, 1994.
- [24] D. J. Thomson and A. D. Chave, "Jackknifed error estimates for spectra, coherences, and transfer functions," in *Advances in Spectrum Analysis and Array Processing*, vol. 1, S. Haykin, Ed. Englewood Cliffs, NJ: Prentice-Hall, 1991, ch. 2, pp. 58–113.
- [25] S. Karnik, J. Romberg, and M. A. Davenport, "Thomson's multitaper method revisited," *IEEE Trans. Inf. Theory*, vol. 68, no. 7, pp. 4864–4891, 2022.
- [26] T. P. Bronez, "On the performance advantage of multitaper spectral analysis," *IEEE Trans. Signal Process.*, vol. 40, no. 12, pp. 2941–2946, 1992.
- [27] A. T. Walden, "A unified view of multitaper multivariate spectral estimation," *Biometrika*, vol. 87, no. 4, pp. 767–788, 2000.
- [28] L. D. Abreu and J. L. Romero, "MSE estimates for multitaper spectral estimation and off-grid compressive sensing," *IEEE Trans. Inf. Theory*, vol. 63, no. 12, pp. 7770–7776, 2017.
- [29] F. J. Harris, "On the use of windows for harmonic analysis with the discrete Fourier transform," *Proc. IEEE*, vol. 66, no. 1, pp. 51–83, 1978.
- [30] P. P. Mitra and B. Pesaran, "Analysis of dynamic brain imaging data," *Biophys. J.*, vol. 76, no. 2, pp. 691–708, 1999.
- [31] H. Bokil, P. Andrews, J. E. Kulkarni, S. Mehta, and P. P. Mitra, "Chronux: a platform for analyzing neural signals," *J. Neurosci. Methods*, vol. 192, no. 1, pp. 146–151, 2010.
- [32] A. Bruns, "Fourier-, Hilbert- and wavelet-based signal analysis: are they really different approaches?" *J. Neurosci. Methods*, vol. 137, no. 2, pp. 321–332, 2004.
- [33] D. J. Thomson, "Quadratic-inverse spectrum estimates: applications to palaeoclimatology," *Philos. Trans. Roy. Soc. London A*, vol. 332, no. 1627, pp. 539–597, 1990.